NPHCDA · UNICEF · GAVI

Predictive Modelling of Zero-Dose and Under-Immunized Children for Improved Immunization Decision Making in Nigeria

Findings and programme use for the National Routine Immunization (RI) team

Prepared by the CIDRE and Quantium Insights LLC consortium, in technical support of NPHCDA, with UNICEF and Gavi (the global vaccine partnership).

Closing the loop on the Executive Director's three requests

All three are complete. Each is answered by slides added or rebuilt since the last review, tagged so they can be checked off one by one.

1 COMPLETE

Nine more antigens, beyond the original four

WHAT YOU ASKED

Add indicators beyond BCG, Penta1, Penta3 and Measles1.

WHAT WAS DELIVERED

Nine antigens added: OPV3, IPV1, PCV3, Yellow Fever, Meningitis A, IPV2, and Rotavirus 1 to 3.

New slide 11

2 COMPLETE

Archetyping rebuilt at local government level

WHAT YOU ASKED

The unsupervised model was at state level. Take archetyping down to LGA.

WHAT WAS DELIVERED

All 774 LGAs grouped into five contextual profiles from 15 local indicators, using no zero-dose or coverage estimate, with candidate packages from each LGA's measured barriers. Two profiles hold 1.52 million missed children.

New slides 20 to 23

3 COMPLETE

The priority states and LGAs, named and ranked

WHAT YOU ASKED

Give the priority states and LGAs for zero-dose focus.

WHAT WAS DELIVERED

All 773 modelled LGAs ranked by burden, each carrying its rate, cumulative share, priority band, equity tier and archetype. 151 LGAs cover half of the burden; 342 cover 80 percent.

New slide 19

How to verify. The six slides answering these requests carry a green tag in the top-right corner: 11, 19, 20, 21, 22 and 23.

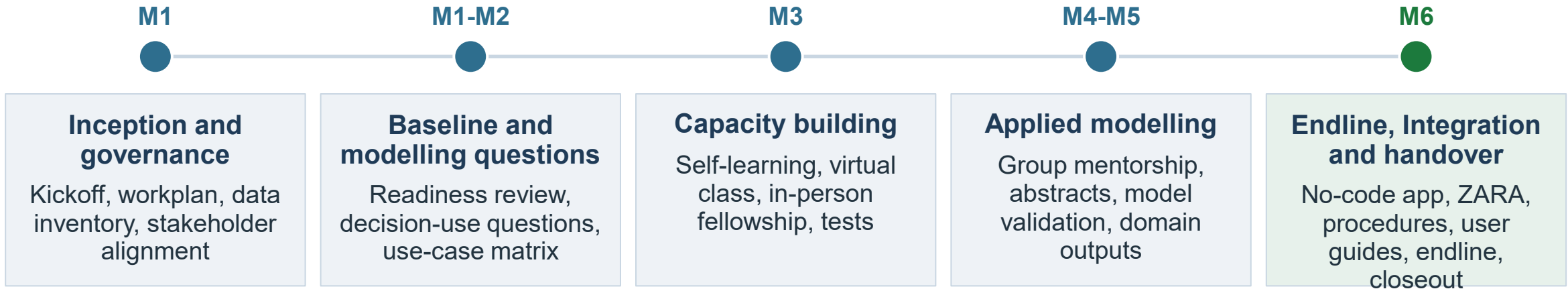
Requests recorded at the Executive Director's review of the first presentation. Prepared by the CIDRE and Quantium Insights LLC consortium, in technical support of NPHCDA.

Outline

1. Project summary: project journey, outputs and handover
2. Key results and programme messages
3. Modelling approach: two Bayesian methods, from state estimates to every LGA
4. External benchmarks and an independent Concordant check
5. Practical use and use cases for decision-making
6. Methodological considerations: data quality, model transparency, archetype robustness
7. The Hugging Face application: filtering, outputs, programme support and limits
8. Recommendations

Project journey: inception to handover

One integrated cycle over six months, from assessment to handover.



Delivery stream

- Baseline assessment - documented status, infrastructure, staff capacity and governance.
- Modelling questions refined into eight domains aligned to NPHCDA decision needs.
- The fellowship combined virtual learning, in-person modelling, and supervised group work.

Evidence of implementation

- Abstract presentations turned model outputs into policy-facing products.
- Endline assessment - measured uptake, readiness gains and remaining barriers.
- Handover packaged models, documentation, dashboards and recommendations.

Key achievement: from one-off training to a modelling workflow NPHCDA now runs in-house.

Source: consortium project record, January to June 2026. DIH = Digital Innovation Hub, the agency's in-house data team; ZARA = Zero-dose Analytics and Risk Assistant.

The seven research questions

Agreed with NPHCDA as the scope of work. Seven domains, one shared question: who is missed, where, and why.

Domain 1

Which antigens are at risk of falling below the 80 percent coverage target in the next 6 to 12 months?

Domain 2

What are the dropout rates across antigen pairs, and what drives incomplete vaccination?

Domain 3

How does zero-dose risk vary by place, wealth, education and access?

Domain 4

How far does vaccine hesitancy explain coverage shortfalls?

Domain 5

Where do zero-dose children concentrate, who are they, and where do we act first?

Domain 6

Where is polio-detection and transmission risk highest on the path to certification?

Domain 7

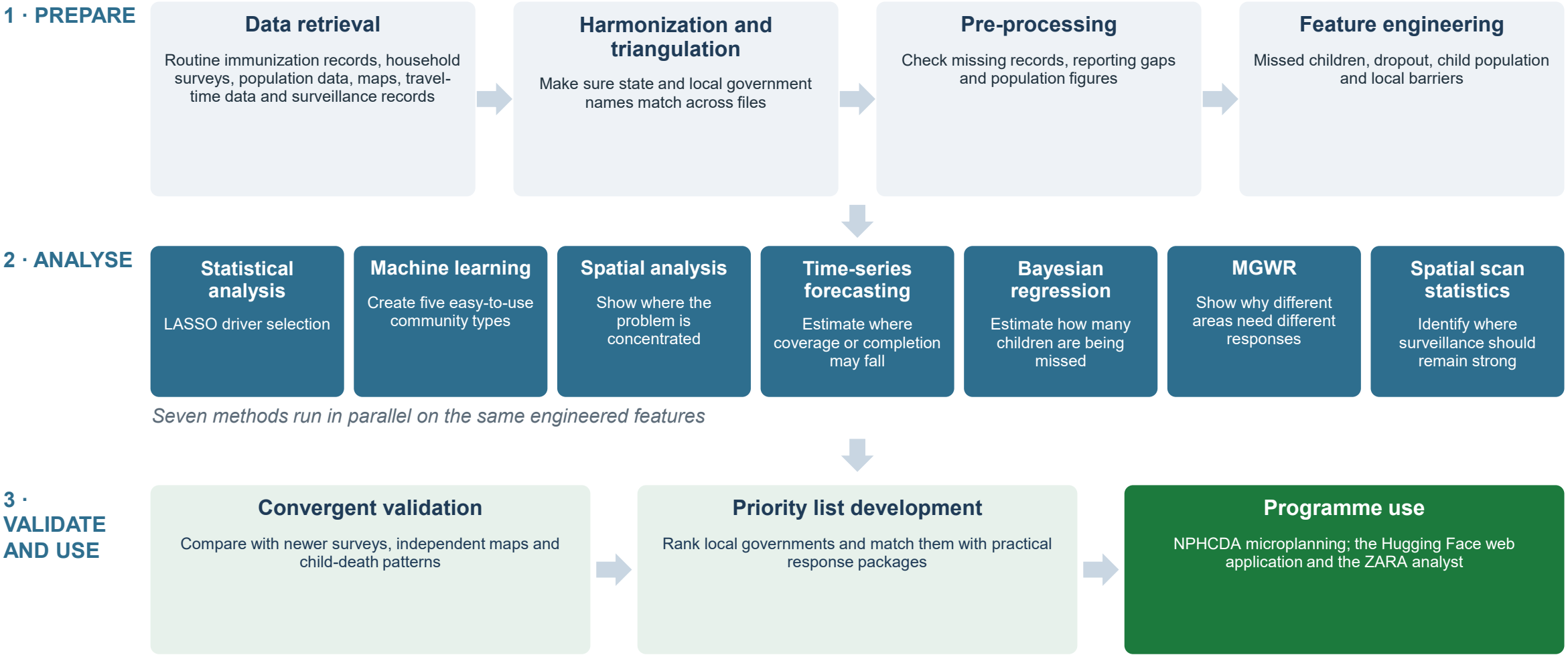
Where is outbreak risk highest, and how does it link to child deaths?

All seven domains are complete, validated and handed over.

Source: scope of work agreed with NPHCDA, 2026. LGA = local government area; RI = routine immunization.

The modelling journey, end to end

We brought many data sources together, checked them, and turned the results into a priority list and simple web tool.



Source: consortium modelling pipeline, 2026. Every acronym is spelled out on the methods, datasets and reference slides.

Methods at a glance

One method per question. Every finding is reproducible from the consortium notebooks.

Domain 1	Showed which vaccines may need attention soon.
Domain 2	Showed where children start vaccination but do not complete it.
Domain 3	Showed where poverty, distance, education and access barriers are strongest.
Domain 4	Checked whether vaccine concerns explain missed children.
Domain 5	Estimated where missed children are (two methods) and grouped areas by barrier.
Domain 6	Showed where polio surveillance and response should remain strong.
Domain 7	Showed where low coverage is linked with higher child-survival risk.

Validation: Independent testing showed the model was highly accurate (3 to 4% error)

Source: consortium methods, 2026. Model estimates reported with their uncertainty; clean convergence, maximum R-hat below 1.01.

The datasets behind every estimate

Eleven sources, mapped to the domains they feed. Nothing is modelled from a single data stream.

Dataset	What it provides	Domains
DHIS2 routine RI reports	Monthly administrative dose counts, all states, 2021 to 2025	1, 2, 5
NDHS 2008 to 2024 (verified)	Survey DTP1 coverage and zero-dose, 12-to-23-month cohort	1, 3, 4, 5
NmDHS 2025-26	New national survey, released after this work: out-of-sample check	Validation
Zero-dose master dataset	37-state file: survey coverage plus 30+ predictors	2, 3, 4, 5
Equity indicators file	Wealth, education, distance to care, urbanicity, fertility	3, 5
GRID3 admin boundaries	Admin-1 and admin-2 layers for maps and contiguity	3, 5, 6
AFP national line list	Case-level surveillance, about 103,000 records, 2015 to 2025	6
IHME GBD 2023 (Nigeria)	State cause-of-death fractions; independent mortality record	7
NPHCDA ACSM register	2,593 advocacy and social-mobilization activities, 36 states	4
Population: NPC 2022 and WorldPop	Local-government denominators for absolute-burden conversion	1, 4, 5, 7
Meta Relative Wealth Index; Weiss travel time; ACLED	Wealth, remoteness and political-violence covariates	3, 5

All acronyms are spelled out in the reference slides.

Local government coverage of the zero-dose estimates

Method 1 estimates 773 of the 774 local government areas; Method 2 estimates all 774.

Zone	LGAs	Method 1	Method 2
North Central	121	121	121
North East	112	111	112
North West	186	186	186
South East	95	95	95
South South	123	123	123
South West	137	137	137
Nigeria	**774**	**773**	**774**

773 / 774

LGAs estimated by Method 1

774 / 774

LGAs estimated by Method 2 (SAE)

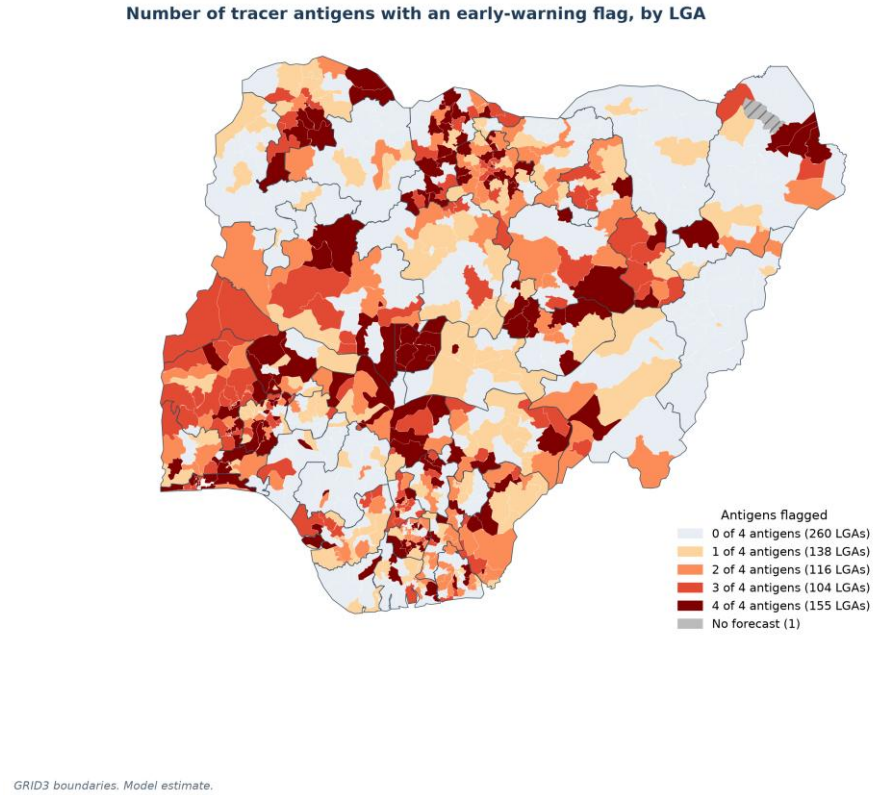
Guzamala (Borno). No Penta1 doses were reported in DHIS2 for 2021 to 2024, and only partially in 2025. Method 1 distributes each state's children using the LGA's share of reported Penta1 doses, so it cannot place Guzamala. Method 2 estimates it from the Borno survey anchor, local conditions and neighbouring LGAs.

What the data team can do. Restore routine Penta1 reporting for Guzamala, and review LGAs whose reported doses are far above or below their estimated child population in the monthly data-quality checks; these are the LGAs where the two methods differ most.

Source: DHIS2 routine antigen export by local government area and month, 2021 to 2025; GRID3 boundaries (774 LGAs). LGA = local government area; SAE = small-area estimation.

Domain 1: which antigens are at risk of falling below target?

An early-warning signal that delivery is falling. It is not a count of children vaccinated.



What to do. Follow up first where the fall is already visible, then watch the projected dips, by vaccine and by LGA.

1,302

early-warning flags, in 513 of the 773 LGAs

3,092 forecasts, one per LGA and tracer vaccine. For 195 flags the fall is already visible in 2025. Nationally, all four stay above their 2024 level.

What below 80 percent means. Doses delivered have dropped below four-fifths of what that area managed in 2024. It does not mean only 80 percent of children were reached.

Model estimate. The 80 percent mark compares each LGA with its own 2024 level of doses delivered. It is a decline tripwire, not coverage of the eligible child population. Map: number of the four tracer vaccines flagged in each LGA. Source: Domain 1 Prophet forecasts on DHIS2 routine data, 2021 to 2025 (Guzamala, Borno, has no 2024 data and no forecast).

No decline warning for the nine additional antigens

Same method, nine more vaccines. The established five hold; rotavirus uptake dipped in 2025 and needs watching

NEW · ED REQUEST 1

5 of 5

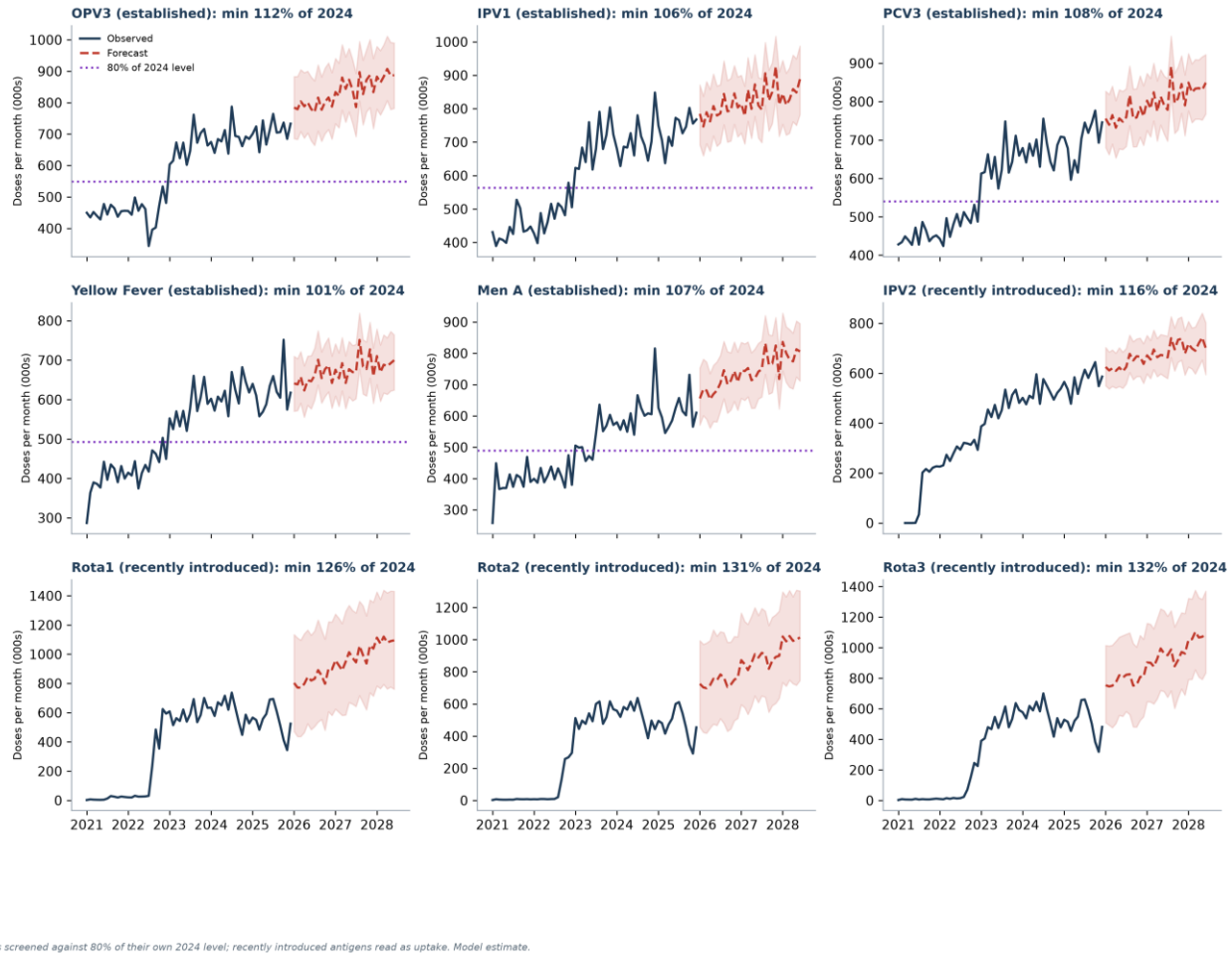
established antigens added - OPV3, IPV1, PCV3, Yellow Fever and Meningitis A - are projected to stay above 80 percent of their own 2024 level. None is flagged.

4 of 4

new series - IPV2 and Rotavirus 1 to 3 - are tracked for uptake. IPV2 is rising; rotavirus was 89-91 percent of 2024 in 2025.

How to read the chart

The purple dotted line is 80 percent of that antigen's own 2024 doses - the at-risk-of-decline early-warning mark. Rising series carry no line, because they should not be screened for decline.



Model estimate. Source: Domain 1 Prophet forecasts on DHIS2 (District Health Information System, version 2) routine data, 2021 to 2025, projected to 2028. OPV = oral polio vaccine; IPV = inactivated polio vaccine; PCV3 = third dose of pneumococcal conjugate vaccine.

What the at-risk-of-decline early-warning measures

It tells you when delivery is slipping against a settled benchmark. It is not a coverage figure.

THE FLAG, IN FOUR LINES

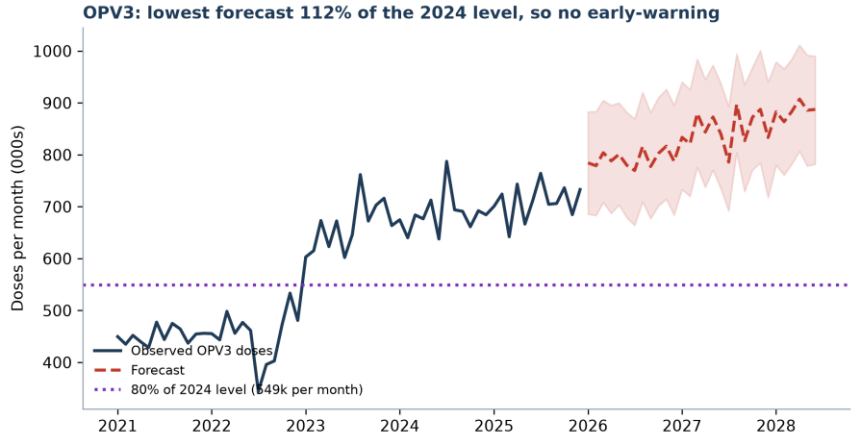
At-risk-of-decline early-warning

What it measures. Projected monthly doses for an antigen, expressed as a percent of that antigen's own 2024 level.

The mark. Below 80 percent means doses are slipping against the settled 2024 benchmark. That is what raises the flag.

It is denominator-free. It does not say what share of children were reached, because no child population enters the calculation.

What to do with it. Catch the decline early, and route supervision to the exact antigen, month and local government area.



Reading one panel. The purple dotted line is 80 percent of OPV3's 2024 doses; the forecast stays clear of it, so OPV3 is not flagged.

Why the benchmark is 2024, not 2025

2024 is the last complete, fully-reported year, so it holds still. 2025 is partial and keeps shifting as late reports arrive, which would raise alerts that are really reporting lag.

Not to be confused with true coverage

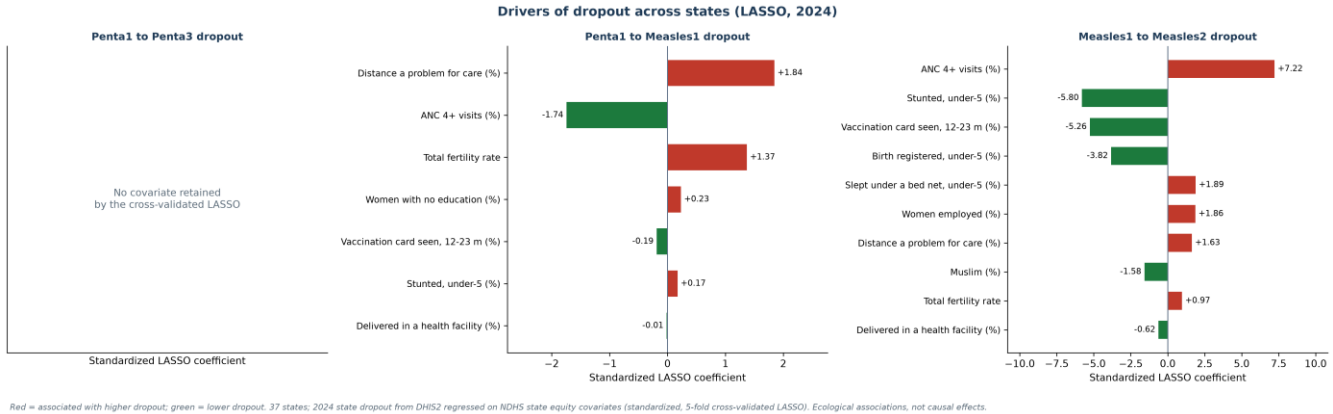
True coverage divides doses by the eligible child population, the 12-to-23-month cohort. It answers a different question and is shown separately in the application as "Estimated coverage (eligible cohort)".

The early-warning tells you when delivery is falling. Coverage tells you how many children you reached. Different questions, different denominators.

Source: Domain 1 methods, consortium documentation, 2026. WHO = World Health Organization; Gavi = the global vaccine partnership. Projected values are model estimates.

Domain 2: what drives incomplete vaccination?

Dropout differs by transition: the second measles dose is still the largest gap, and each transition has different drivers.



Red = associated with higher dropout; green = lower dropout. 37 states; 2024 state dropout from DHIS2 regressed on NDHS state equity covariates (standardized, 5-fold cross-validated LASSO). Ecological associations, not causal effects.

The biggest gap is between Measles1 and Measles2: national dropout fell from 60% in 2021 to 34% in 2025, still far above Penta1 to Penta3 (about 5%) and Penta1 to Measles1 (about 11%).

Penta1 to Penta3

Higher with conflict deaths; lower with ORS use (LGAs).

Defaulter tracing and security-sensitive outreach in conflict-affected LGAs.

Penta1 to Measles1

Distance to care; high fertility; less antenatal care.

Link the 9-month visit to outreach and nutrition contacts.

Measles1 to Measles2

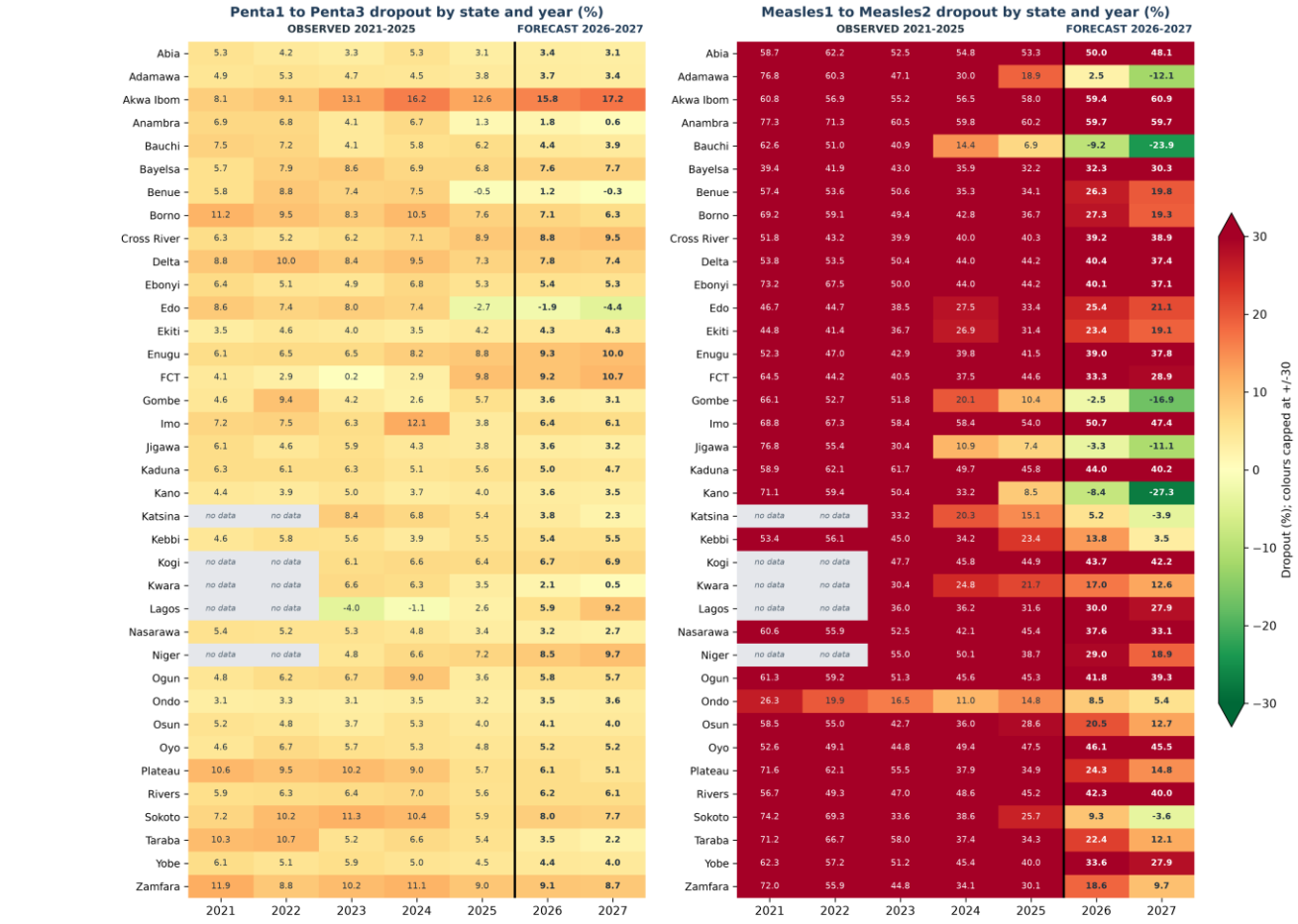
Few vaccination cards seen; births unregistered.

Home-based records, card retention and second-year reminders.

Model estimate. DHIS2 2021-2025 (counts read with thousands separators). Cross-validated LASSO: 37 states (2024; chart) and 743 LGAs (2021-2024). Ecological associations: directional, not causal. ORS = oral rehydration salts.

Domain 2: where dropout is highest, by state

Observed 2021-2025 and Prophet forecast 2026-2027, the same view as the web application's heatmap.



Observed: mean of monthly state dropout from DHIS2 (counts parsed with thousands separators). Grey = no DHIS2 records for the state that year. Forecast: calendar-year mean of monthly Prophet forecasts per state. Negative values mean the later dose was recorded at or above the earlier dose.

34%

Measles1 to Measles2 dropout nationally in 2025 (60% in 2021)

34 of 37

states lost more than 10% of children between the two measles doses in 2025

about 5%

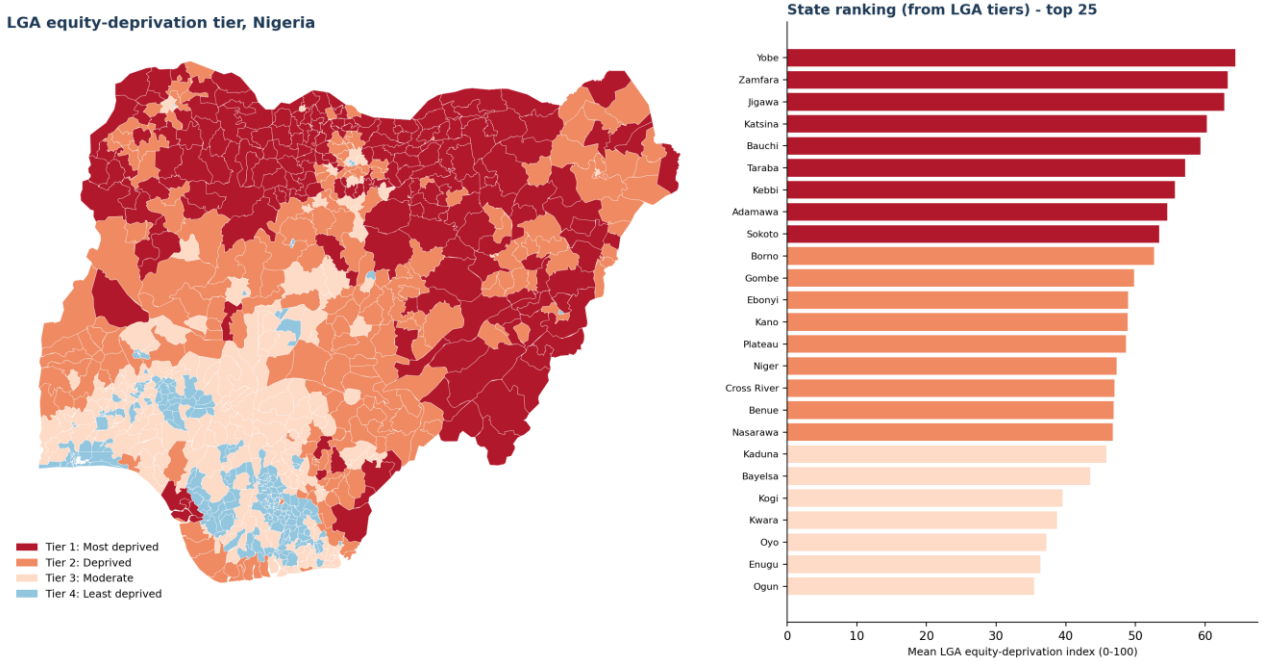
Penta1 to Penta3 dropout in 2025; low and stable since 2021

Act first in Anambra, Akwa Ibom, Imo, Abia and Oyo, where 48-60% of children who receive Measles1 miss Measles2.

Model estimate. DHIS2 monthly counts read with thousands separators; grey = no records for the state that year; negative dropout = later dose recorded at or above the earlier dose.

Domain 3: where children are hardest to reach

Model estimate. The index combines travel time, education, poverty and wealth.



Tier 1 Most deprived

Tier 2 Deprived

Tier 3 Moderate

Tier 4 Least deprived

Darker areas have stronger barriers: longer travel, lower education and more poverty.

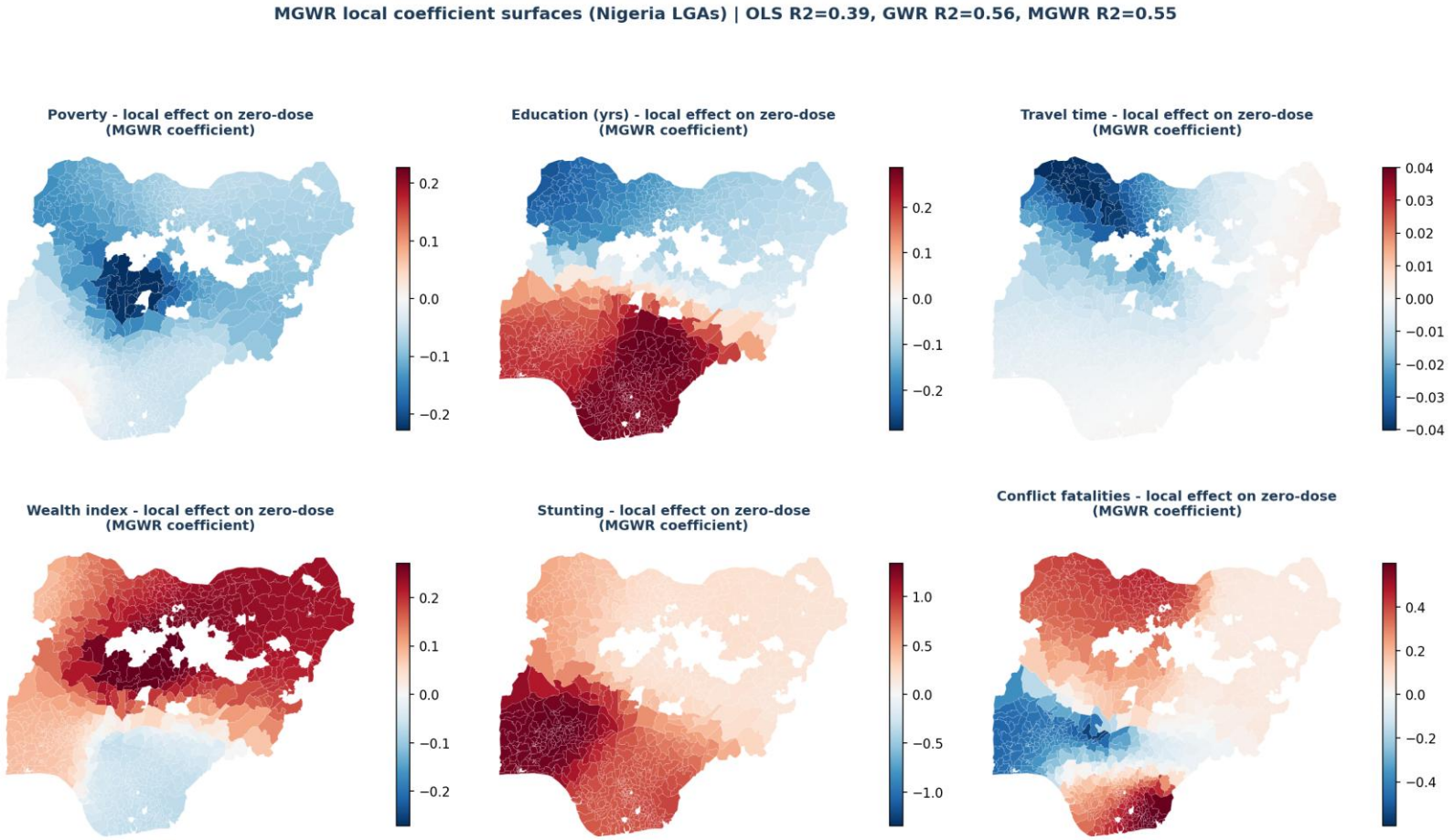
Many of the hardest-to-reach LGAs are in the North-East and North-West.

Use this with the burden ranking. Burden shows where the most children are; this tier shows where barriers are deepest. Areas with both high burden and high deprivation should be prioritized..

Model estimate. Composite equity-deprivation index (0 to 100) built from Weiss travel time, education, poverty and the Meta Relative Wealth Index, grouped into four equal bands; GRID3 boundaries.

The causes act locally

The previous slide showed where disadvantage is deepest. These maps show which barrier appears strongest in each place.



39% → 55%

of the variation explained, once drivers can act differently in different places

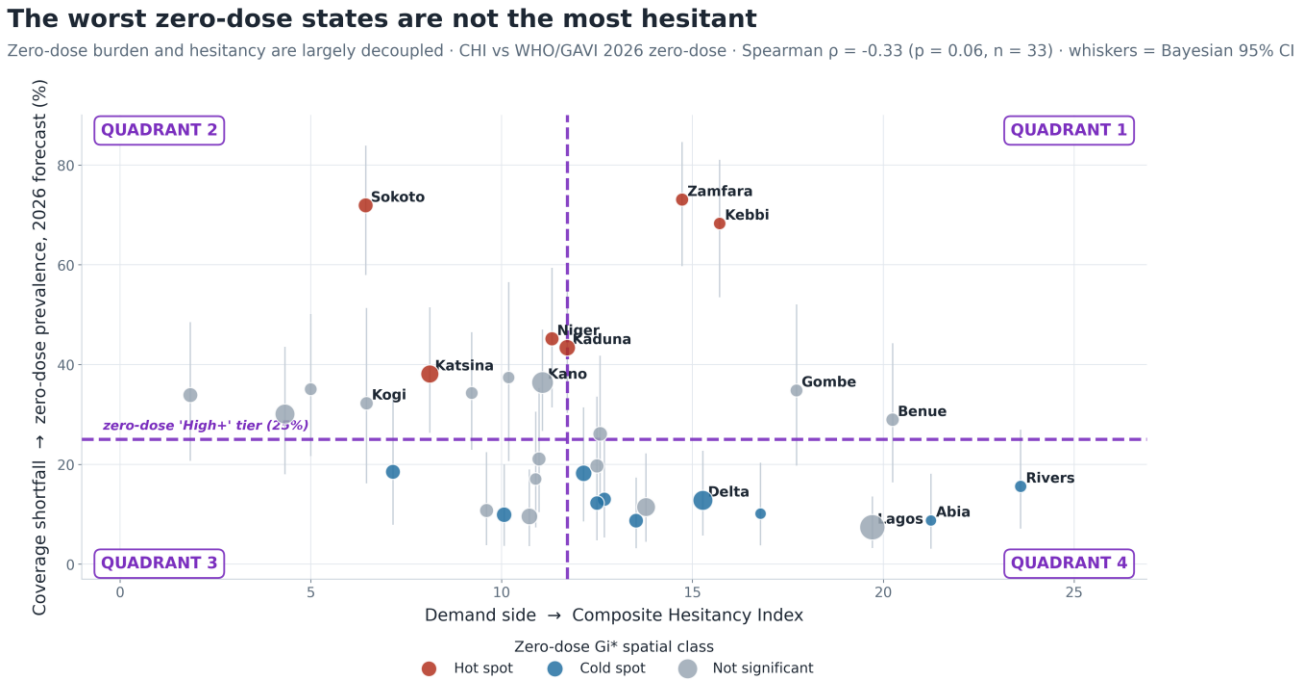
Each map shows one possible barrier to vaccination.

Red means the factor is linked with more zero-dose children. Blue means the opposite. Darker colour means stronger effect.

Model estimate across the 774 local government areas. MGWR = multiscale geographically weighted regression, which lets each driver's effect vary from place to place. These are local, standardised, partial associations, not causal effects. White areas are not modelled because they have no DTP1 reporting, so they carry no estimate rather than a zero.

Domain 4: does hesitancy explain the shortfall?

Hesitancy is real, but it does not sit where the missed children are.



rho = -0.33

The states with the most recorded doubt are not the states missing the most children.

Each bubble is a state. Bubble size is the under-five population.

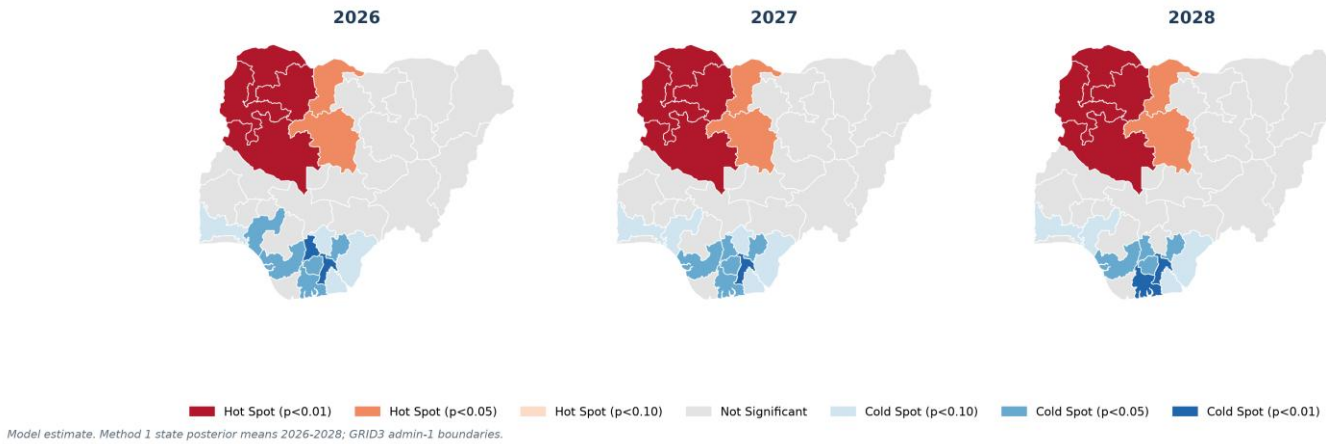
Source: NPHCDA ACSM monitoring register (acsm_cleaned.xlsx), 2,593 activities, 36 states. Analysis: Domain 4, NPHCDA/UNICEF Immunization Modelling Framework. Zero-dose: Bayesian hierarchical forecast (NDHS 2008–2024 → 2026), DTP1-based WHO/GAVI definition.

What to do. In high zero-dose but low-hesitancy areas, focus on access, outreach and service availability. Use trust-building where hesitancy is actually high.

Model estimate. Domain 4 Composite Hesitancy Index against the Domain 5 zero-dose forecast. rho = Spearman rank correlation; full provenance on the figure.

Domain 5: where do zero-dose children concentrate?

A North-West hotspot that stays fixed from 2026 to 2028, held in place by structural conditions.



2.10 million

zero-dose children in 2026 (95% interval 1.90 to 2.31 million), across 773 modelled LGAs

51%

of the burden is in the top 20% of LGAs (155 LGAs)

The maps show the modelled zero-dose surface for 2026, 2027 and 2028. The hot area sits over the North-West in every year.

Sokoto, Kebbi and Zamfara remain high-priority throughout the forecast period. They require a multi-year plan, not a one-off activity.

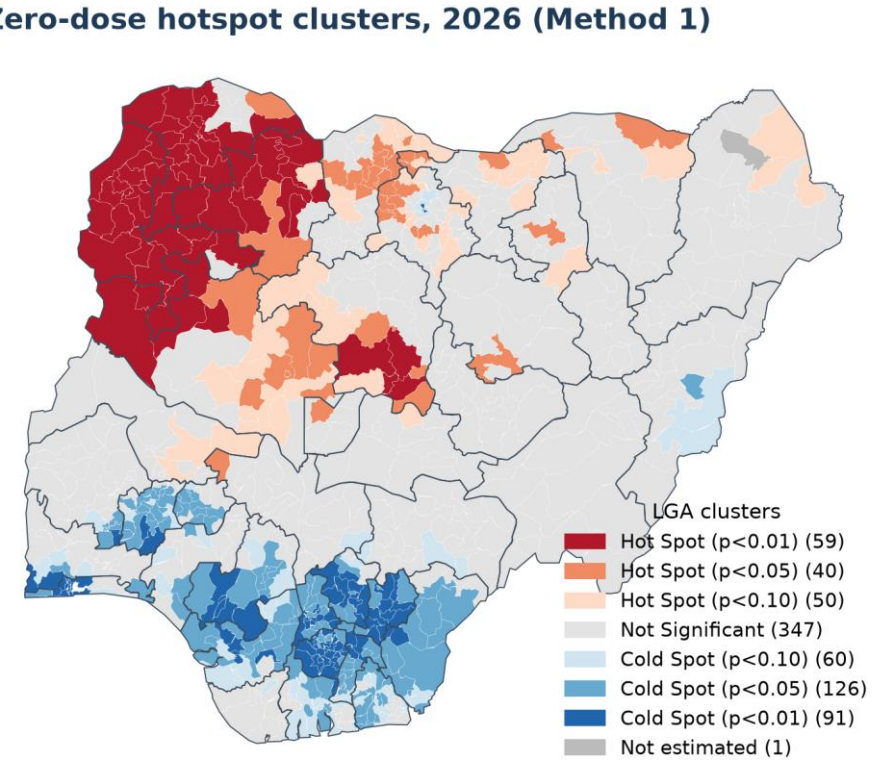
342 local government areas hold 80 percent - concentrated, but not the textbook 80/20.

Model estimate. Source: Method 1: Bayesian hierarchical model with DHIS2-calibrated LGA allocation (Bayesian hierarchical Beta regression on NDHS 2008 to 2024 with a DHIS2 trend term; LGA allocation by DHIS2 Penta1 share and NPC 2022 population); GRID3 boundaries; Getis-Ord Gi state hotspots (Queen contiguity).*

NEW · ED REQUEST 3

Every local government area, ranked by burden

The top 151 reach half of zero-dose children; the top 342 reach 80 percent.



Red marks clusters of high zero-dose burden; blue marks low-burden clusters.

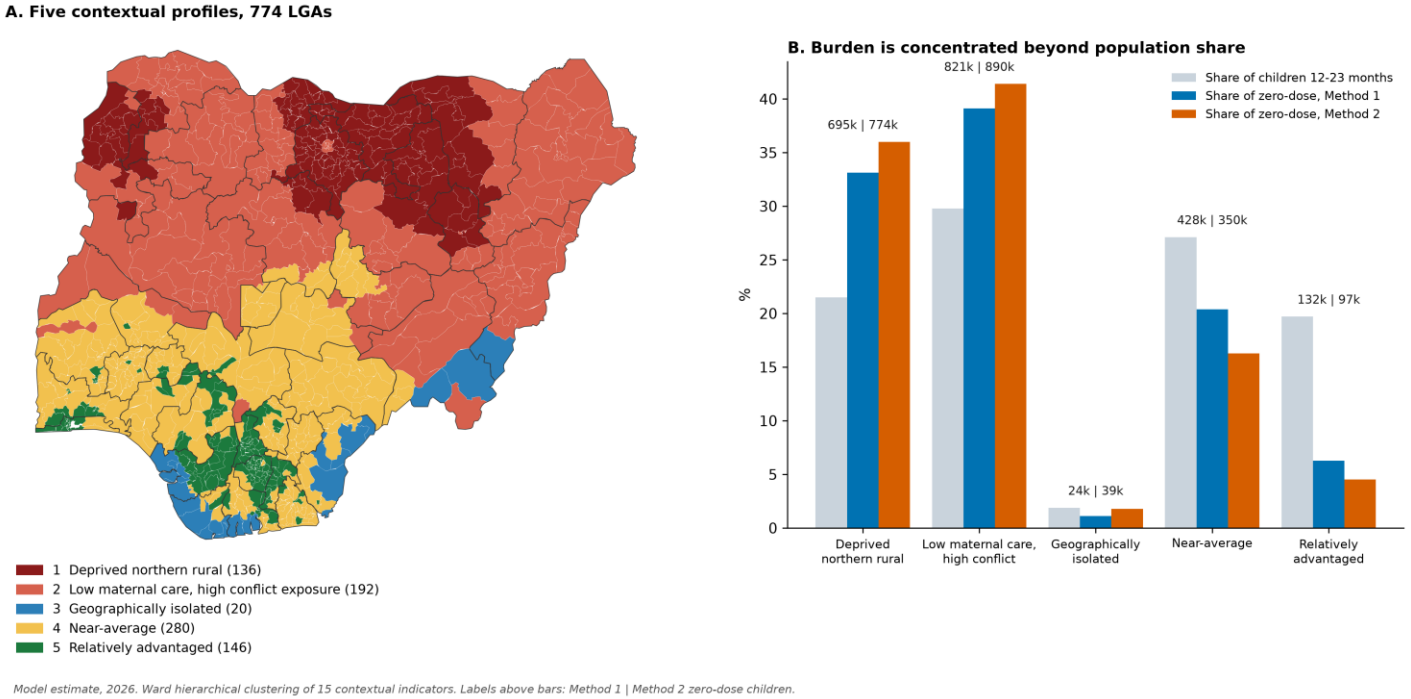
Rank	State	LGA	Zone	Burden	Rate %
1	Zamfara	Bukkuyum	North West	13,589	87.8
2	Zamfara	Bakura	North West	13,253	99.0
3	Sokoto	Tambuwal	North West	12,738	83.2
4	Zamfara	Birnin Magaji	North West	12,223	92.8
5	Zamfara	Maradun	North West	12,057	81.2
6	Zamfara	Maru	North West	12,052	57.5
7	Zamfara	Zurmi	North West	11,913	56.6
8	Sokoto	Sokoto North	North West	11,869	75.2
9	Zamfara	Talata Mafara	North West	11,666	75.6
10	Zamfara	Kaura Namoda	North West	11,282	55.3

Top 10 shown here (Method 1). The full ranked list for both methods is in the government workbook and can be filtered by state or priority level in the web app.

Model estimate. Source: Method 1: Bayesian hierarchical model with DHIS2-calibrated LGA allocation, 2026; GRID3 boundaries. Hotspot method: Getis-Ord Gi (spatial clustering), k = 5 nearest neighbours. Rates are capped at 99%.*

NEW · ED REQUEST 2

Archetype profiles and modelled zero-dose burden



Model estimate, 2026. Ward hierarchical clustering of 15 contextual indicators. Labels above bars: Method 1 | Method 2 zero-dose children.

Built from 15 contextual indicators (education, nutrition, poverty, wealth, travel time, water, maternal care, conflict and DTP1-3 dropout), the profiles use no zero-dose or coverage estimate. Yet they reproduce where the children are: profiles 1 and 2 hold 1.52 million of 2.10 million zero-dose children (72 percent; 77 percent under Method 2).

Colour shows the contextual profile. Bars compare each profile's share of children with its share of zero-dose children under both methods. Labels describe only what the indicators measure.

Model estimate. Profiles (community archetypes) from agglomerative (Ward) clustering of 15 LGA indicators (IHME, DHS, Meta Relative Wealth Index, Weiss travel time, ACLED); bars are modelled zero-dose children under both methods; GRID3 admin-2 boundaries.

NEW · ED REQUEST 2

Five contextual profiles and candidate responses

Profiles 1 and 2 hold about 72 percent of all zero-dose children (77 percent under Method 2).

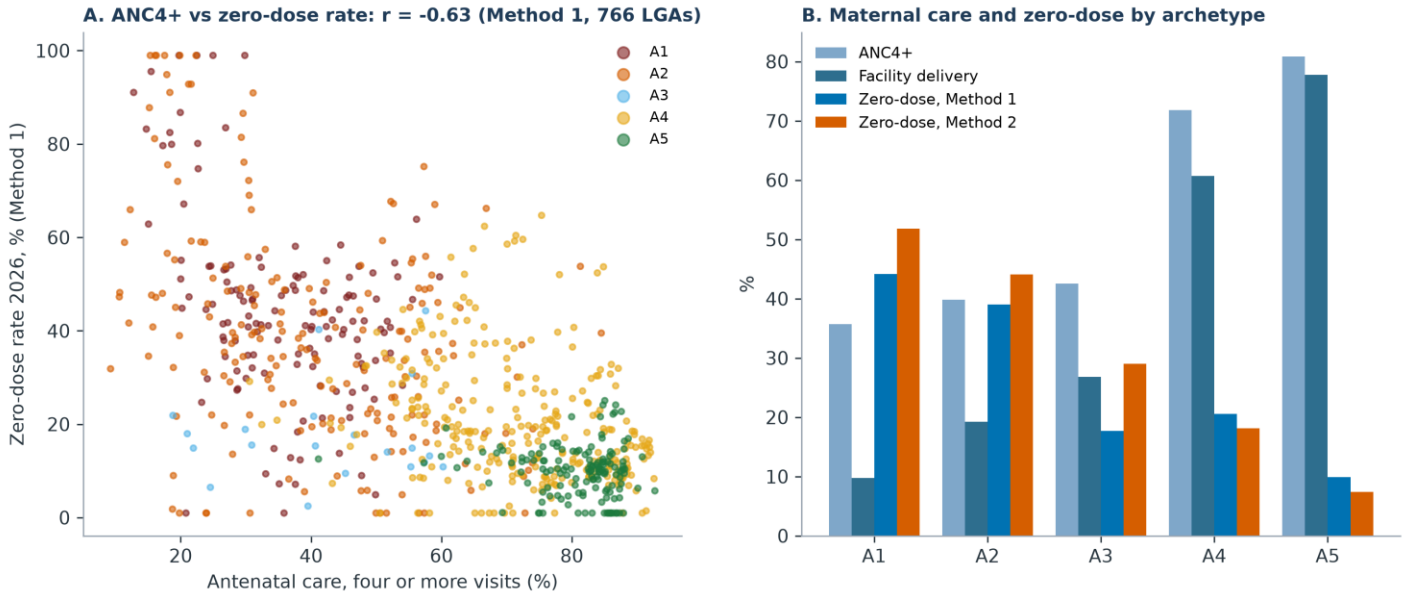
#	Profile	LGAs	Zero-dose children	Mean rate	Defining signature	Priority intervention
1	**Deprived northern rural** Kano, Jigawa, Katsina	136	**695,337 (33%)**	44.2%	Highest child undernutrition; lowest maternal education (about 2.5 years) and facility delivery (about 10%)	Reaching-Every-Community microplanning, BHCPF outreach and community health workers; co-deliver with nutrition and antenatal care
2	**Low maternal care, high conflict exposure** Borno, Niger, Adamawa	192	**820,871 (39%)**	39.0%	Highest political-violence fatalities; high antigen dropout; weak antenatal care and delivery	Security-integrated microplanning, negotiated access, mobile outreach teams and defaulter tracing
3	**Geographically isolated** Cross River, Bayelsa, Delta	20	**23,514 (1%)**	17.7%	Extreme travel time to care (about 108 minutes); low improved-water access	Outreach and mobile sessions (boat-based where waterways apply), community health workers, multi-antigen bundling
4	**Near-average** Osun, Akwa Ibom, Oyo	280	**427,791 (20%)**	20.6%	Near the national average on every indicator; fair care-seeking (antenatal care 72%, delivery 61%)	Core fixed-site and outreach services, ward enumeration, and Periodic Intensification of Routine Immunization
5	**Relatively advantaged** Imo, Anambra, Lagos	146	**131,691 (6%)**	9.9%	Highest wealth and education (about 9.6 years) and facility delivery (78%); moderate conflict events	Targeted social mobilisation, a fixed-plus-outreach hybrid, and private-sector last-mile reach

Model estimate. Ward clustering of 15 indicators across all 774 LGAs; labels describe measured indicators only. Zero-dose children and mean LGA rate from Method 1 (total 2.10 million). Grey text names example states. BHCPF = Basic Health Care Provision Fund.

Maternal and immunization gaps sit in the same places

NEW · ED REQUEST 2

The local governments with the weakest antenatal care and facility delivery are the same ones missing the most zero-dose children, so an existing maternal contact is a ready entry point to reach them.



Model estimate. n = 767 LGAs with maternal-care data (766 with a Method 1 estimate). DHS Spatial Data Repository 2018 surfaces.

1.5 million

of the 2.10 million modelled zero-dose children (Method 1) are in the two profiles with the weakest maternal care.

r = -0.63

antenatal care, four or more visits, against the Method 1 zero-dose rate across local governments. Facility delivery is r = -0.59.

The link. Weaker antenatal care goes with higher zero-dose across the local governments with antenatal-care data (ANC4+ r = -0.63; facility delivery r = -0.59). The link is strong and consistent.

The burden. The deprived northern rural and low maternal care, high conflict exposure profiles hold about 1.5 million zero-dose children, on the lowest antenatal care (36 to 40%) and facility delivery (10 to 19%).

The action. Co-deliver immunization with antenatal and delivery contacts in these profiles: one visit, several services, rather than immunization outreach on its own.

Model estimate. Source: LGA profile master (IHME and DHS antenatal-care and facility-delivery covariates; modelled zero-dose). RMNCH = reproductive, maternal, newborn and child health; EPI = Expanded Programme on Immunization; ANC4+ = antenatal care, four or more visits; LGA = local government area.

From profile to action: the RMNCH-EPI operational list

NEW · ED REQUEST 2

Five contextual profiles, the maternal-care numbers behind each, and the candidate package.

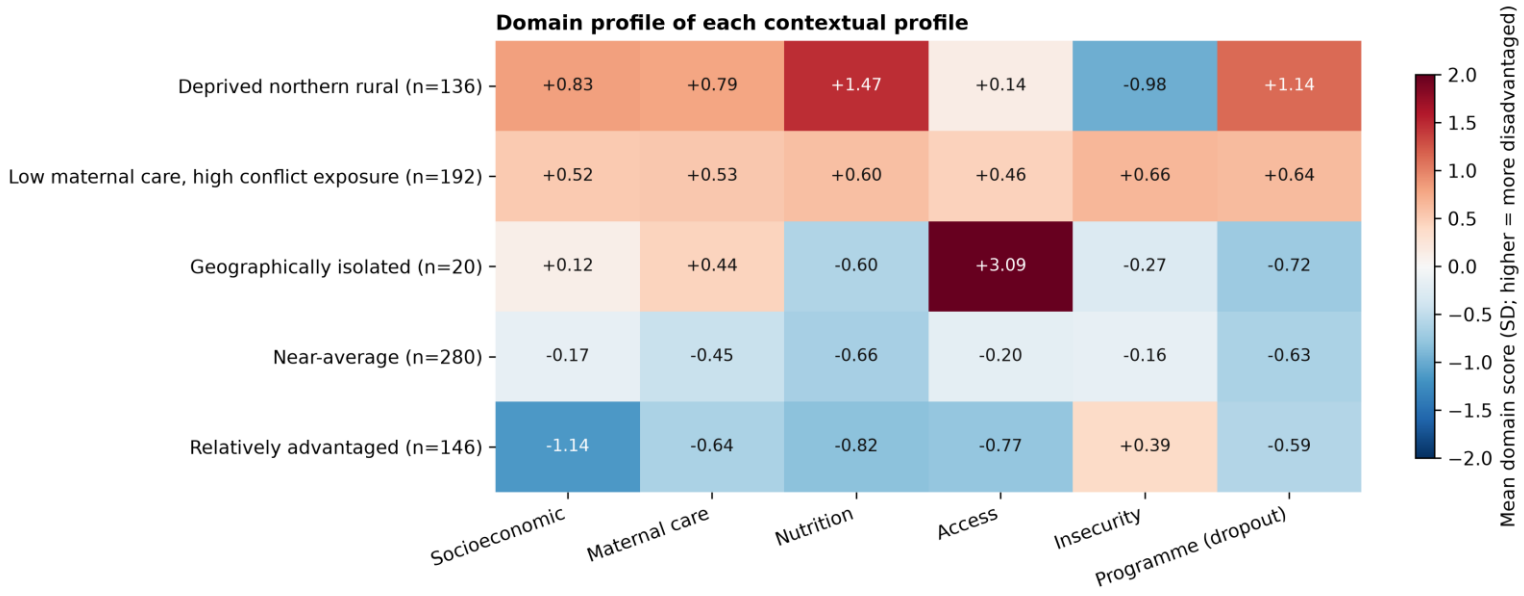
Profile	LGAs	ANC4+ (%)	Facility delivery (%)	Zero-dose rate (%)	Zero-dose children	RMNCH-EPI intervention, matched to the profile
Deprived northern rural	136	35.7	9.8	**44.2**	**695,337**	Reaching-Every-Community microplanning, BHCPF outreach and community health workers; co-deliver immunization with antenatal and nutrition contacts
Low maternal care, high conflict exposure	192	39.8	19.2	**39.0**	**820,871**	Security-integrated microplanning, negotiated access, and mobile outreach teams delivering immunization and antenatal care together
Geographically isolated	20	42.6	26.8	**17.7**	**23,514**	Outreach and mobile sessions (boat-based where waterways apply), community health workers, and multi-antigen and antenatal bundling per visit
Near-average	280	71.8	60.7	**20.6**	**427,791**	Ward enumeration and Periodic Intensification of Routine Immunization linked to antenatal clinics
Relatively advantaged	146	80.9	77.8	**9.9**	**131,691**	Targeted social mobilization, a fixed-plus-outreach hybrid, and private-sector last-mile reach for immunization and antenatal care

The full list. All 774 local government areas are ranked worst-to-best by modelled zero-dose burden in the operational workbook, each carrying its profile, maternal-care figures and candidate package, ready for microplanning.

Model estimate. Source: LGA archetype master and the RMNCH-EPI operational workbook (IHME and DHS antenatal-care and facility-delivery covariates; Method 1 modelled zero-dose). BHCPF = Basic Health Care Provision Fund. Facility-level primary health care lists and facility density are not included; those need a GRID3 or National Health Facility Registry dataset.

Five contextual profiles: what sets them apart

Mean domain scores (higher = more disadvantaged); labels describe only what the indicators measure.



Domain score = mean signed z of its indicators. Access includes travel time and improved water; programme = DTP1-3 dropout.

An operational choice. Four profiles fit slightly better (silhouette 0.31 vs 0.18); five are kept because near-average and relatively advantaged LGAs differ in burden (16.3% vs 4.5%).

Coherent. 79% of neighbouring LGAs share a profile (26% by chance); bootstrap stability ARI 0.54 (0.42 to 0.72).

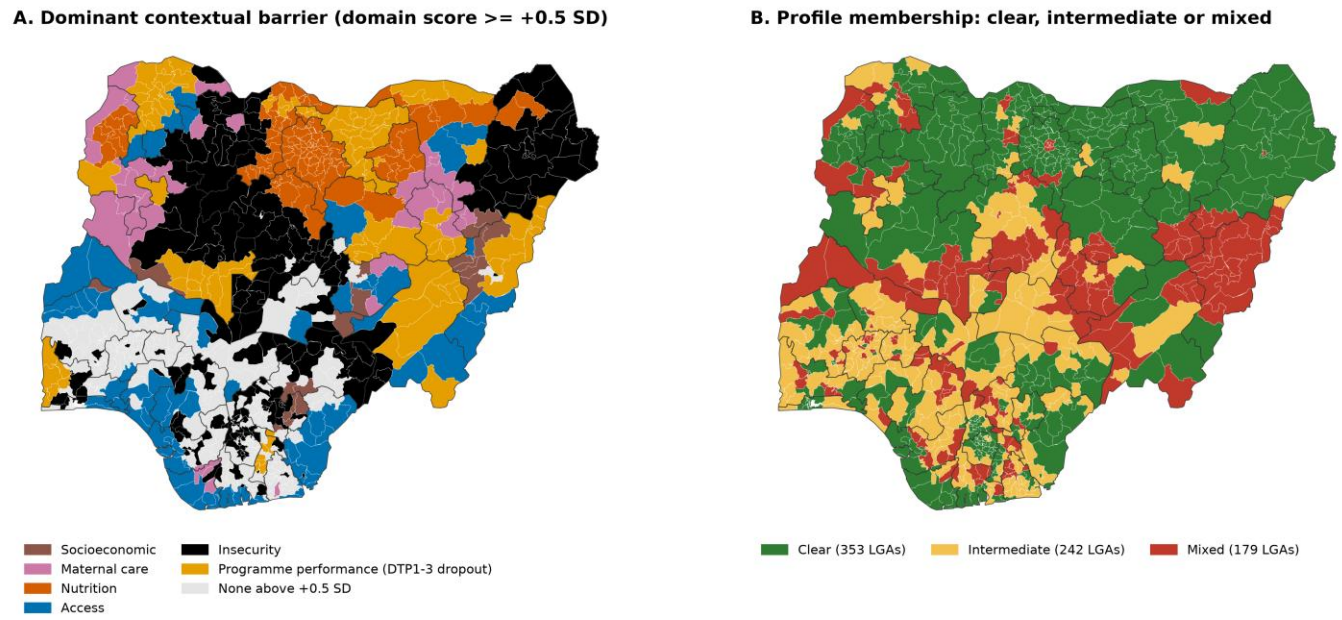
Membership. Clear for 46% of LGAs (60% of modelled zero-dose children), intermediate for 31%, mixed for 23%.

Labels retired. Earlier names for nomadic, migrant, riverine and urban-slum areas were dropped because no indicator measures those attributes. Population mobility must be confirmed locally before transit-point strategies are used. The two highest-burden profiles hold 77.4% of Method 2 modelled zero-dose children but 51.3% of the birth cohort.

Model estimate. Ward hierarchical clustering of 15 indicators in six domains (socioeconomic, maternal care, nutrition, access, insecurity, DTP1-3 dropout), 774 LGAs. ARI = adjusted Rand index; SD = standard deviation.

Barriers differ within states and at the same zero-dose rate

Dominant barrier by LGA, and how clearly each LGA fits its profile.



Clear: bootstrap assignment frequency >= 0.8 and centroid distance ratio < 0.8; mixed: frequency < 0.5 or ratio >= 0.9; intermediate: all others. These are stability and affinity measures, not probabilities.

35 of 37

states contain LGAs with different dominant barriers

56% / 59%

of access and insecurity variance lies within states (nutrition 5%)

67%

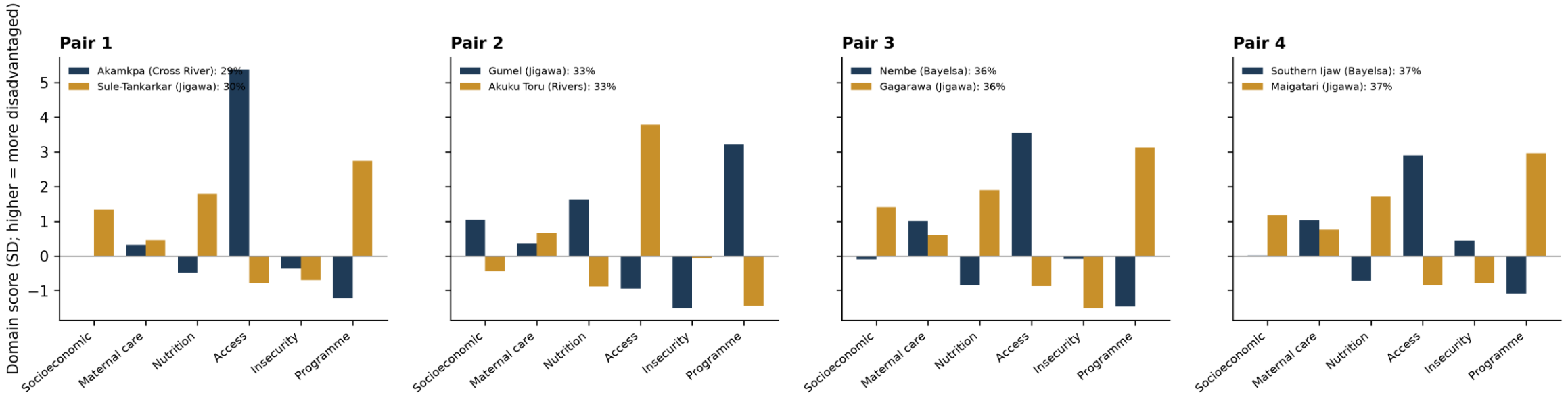
of LGA pairs with near-equal rates have a different dominant barrier

Caveat. Near-equal pairs are still contextually closer than random pairs (median distance 1.8 vs 2.8), so 'same rate, different barriers' applies to a substantial minority, not to every LGA. State averages capture nutrition and maternal care well but hide access and insecurity, which need LGA-level planning.

Model estimate. Ward hierarchical clustering of 15 indicators in six domains (socioeconomic, maternal care, nutrition, access, insecurity, DTP1-3 dropout), 774 LGAs. ARI = adjusted Rand index; SD = standard deviation. Dominant barrier = highest domain score at or above +0.5 SD.

Same estimated rate, different barriers

Four real LGA pairs with near-identical Method 2 zero-dose rates.



Pairs with near-identical SAE 2026 rates (intervals overlap almost entirely) but different dominant barriers. Real LGAs (B_R3).

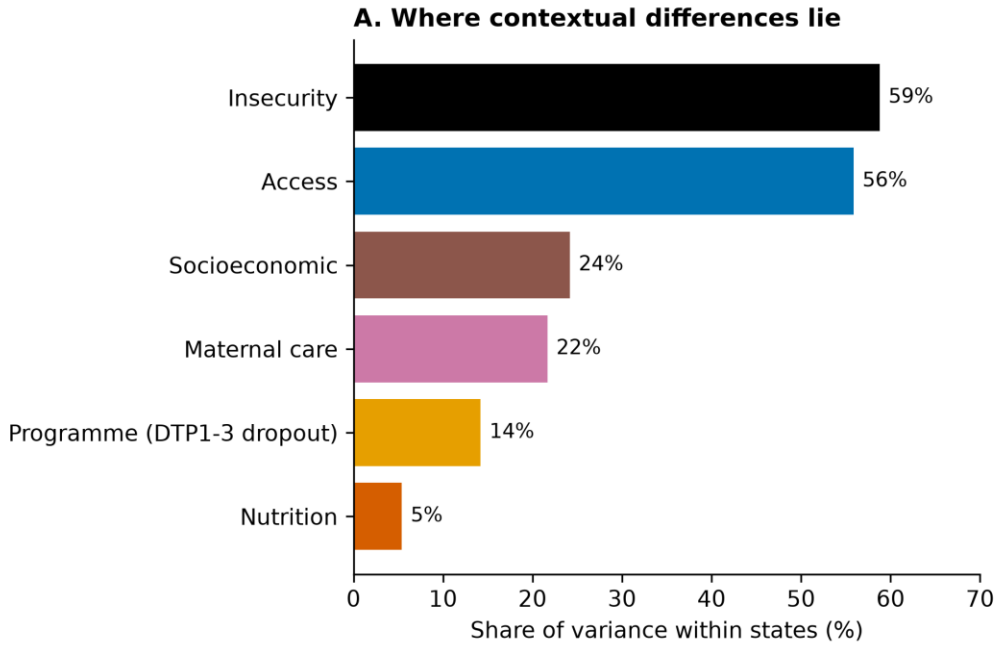
Reading the pairs. Each pair has nearly the same estimated zero-dose rate but a different dominant barrier, so a rate-only ranking would send both the same response. Across all near-equal pairs, 49.6% fall in different profiles.

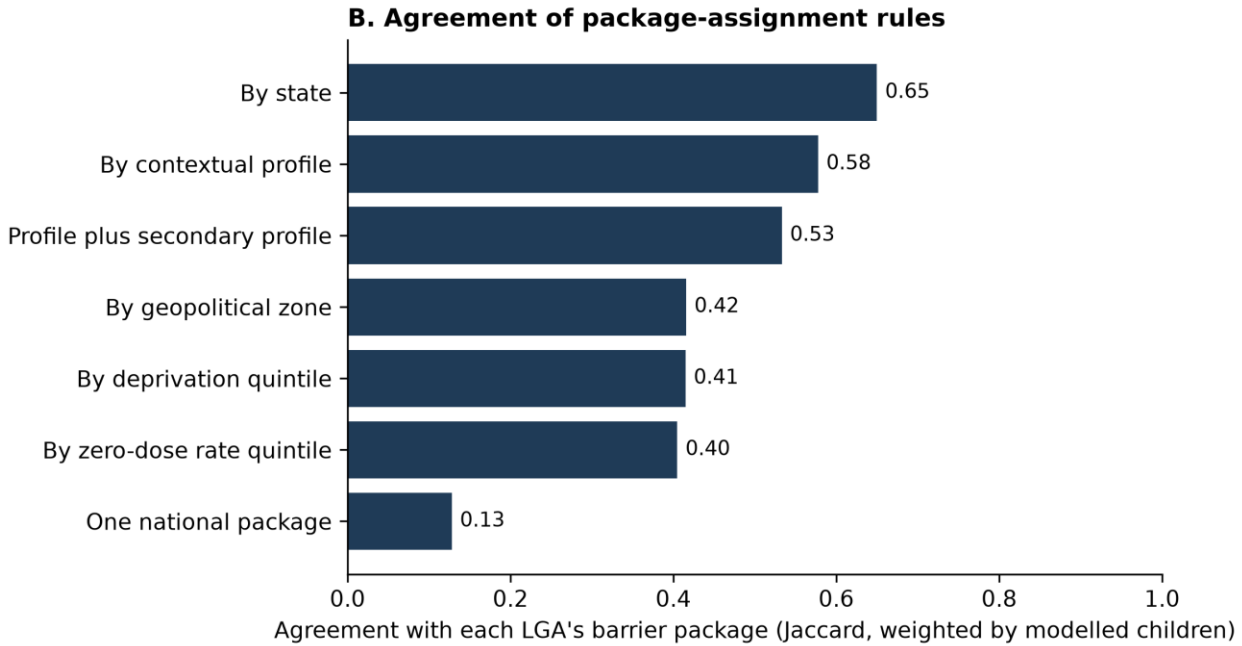
Caution. All four pairs contrast Niger Delta with Jigawa LGAs, and their rate intervals are wide. They illustrate the pattern; they do not measure how common it is.

Model estimate. Ward hierarchical clustering of 15 indicators in six domains (socioeconomic, maternal care, nutrition, access, insecurity, DTP1-3 dropout), 774 LGAs. ARI = adjusted Rand index; SD = standard deviation.

Assigning packages: start from the state, refine by LGA barriers

Agreement of each assignment rule with every LGA's own barrier package, weighted by modelled zero-dose children.





Packages are candidate component sets from documented rules; agreement measures consistency with the rules, not intervention effectiveness.

Combined barriers are the norm. 36% of LGAs, holding 67% of modelled zero-dose children, have two or more flagged barriers.

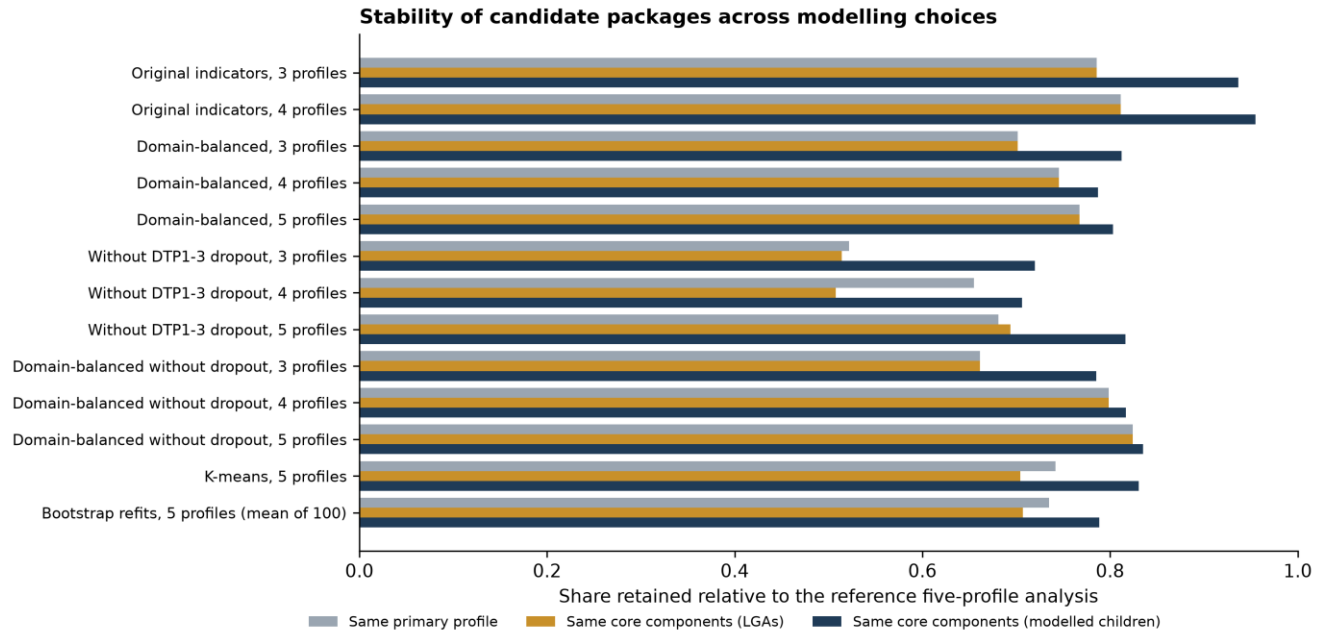
State first, then LGA. State packages agree best (0.65), profiles next (0.58); one national package agrees least (0.13).

Validate locally. Packages are candidate components from documented rules, not measured intervention effects.

Model estimate. Barrier flag = domain score in the national top quartile (travel time: top decile). Agreement = Jaccard similarity (1 = identical). Population mobility is not measured and is never assigned.

How robust are the candidate packages?

Share of LGAs and children keeping the same core components when modelling choices change.



Core components = Jaccard similarity >= 0.67. LGA-level barrier flags do not depend on clustering; changing the flagging threshold retains 73-82%.

Stable core. Core components are kept for 51-82% of LGAs (71-96% of modelled children) across indicator sets, numbers of profiles, algorithms and bootstrap refits.

Threshold. Changing the barrier-flagging threshold keeps 73-82% of LGA packages; LGA flags do not depend on the clustering.

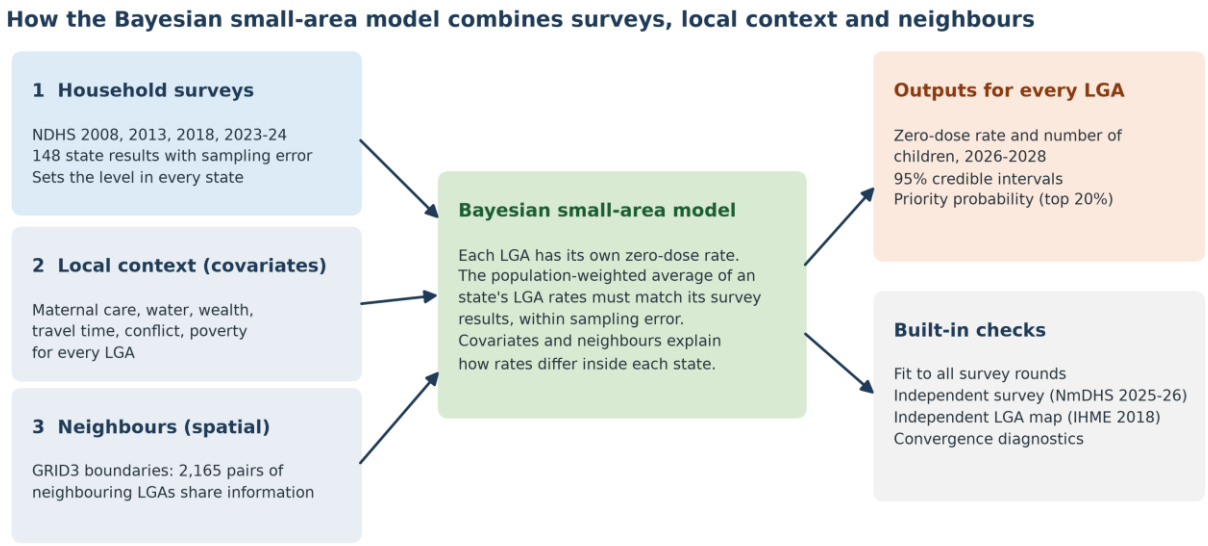
Information content. Profiles add to zone in explaining LGA zero-dose (R-squared +0.08) but little beyond a deprivation quintile (+0.01); at state level they improve prediction of the NmDHS 2025-26 (R-squared 0.71 to 0.78).

Use. Choose components from LGA barrier flags; use profiles as a planning summary, not as an independent predictor.

Model estimate. Ward hierarchical clustering of 15 indicators in six domains (socioeconomic, maternal care, nutrition, access, insecurity, DTP1-3 dropout), 774 LGAs. ARI = adjusted Rand index; SD = standard deviation. Core components = Jaccard similarity of 0.67 or more; R-squared out of sample (state check leave-one-out).

Method 2: Bayesian small-area estimation (SAE)

A second, survey-anchored model that estimates zero-dose directly for every one of the 774 LGAs.



All 774 LGAs. Inference by Markov chain Monte Carlo (4 chains x 8,000 draws).

1. **Anchored to the surveys.** Each state's LGA estimates must add up to that state's NDHS zero-dose level (2008, 2013, 2018, 2023-24), allowing for survey sampling error.

2. **Local conditions.** Six LGA measures shape differences within a state: maternal care, improved water, wealth, travel time to a facility, conflict and poverty.

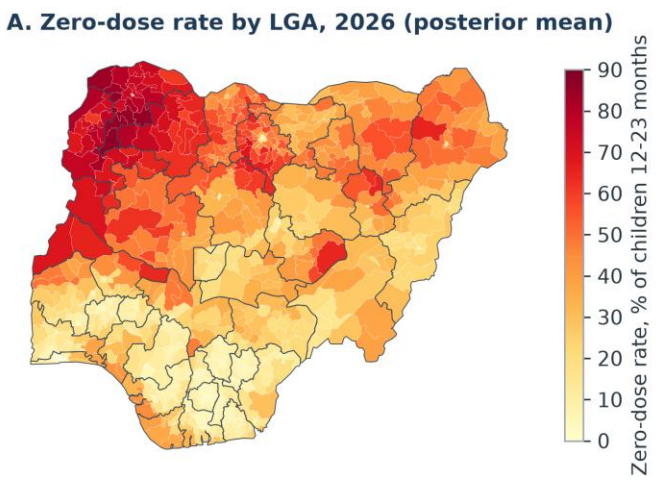
3. **Neighbours.** A spatial term (BYM2) lets neighbouring LGAs share information, so sparse areas borrow strength from their surroundings.

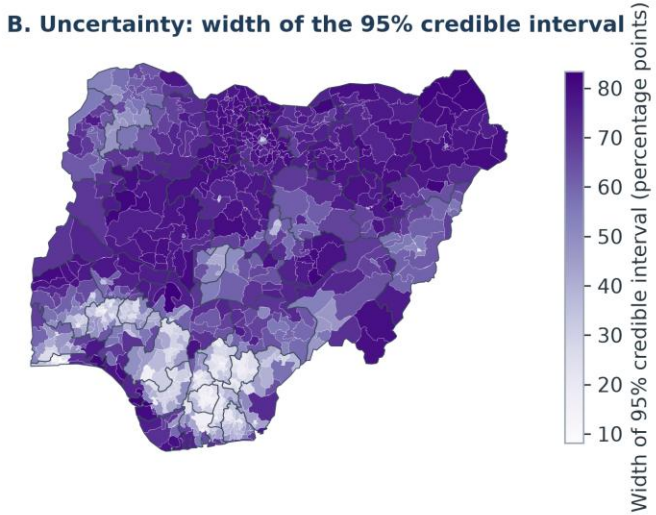
Why a second method. Method 1 distributes each state's estimate to its LGAs using routine DHIS2 Penta1 reporting. Method 2 does not use routine coverage at all: it combines survey levels with local conditions and geography, and gives every LGA its own 95% credible interval and a probability of being a priority. Reading both together shows which priorities are robust.

Model estimate. NDHS = Nigeria Demographic and Health Survey; BYM2 = Besag-York-Mollie spatial model, version 2. Fitted in PyMC (4 chains x 8,000 draws). Covariates: DHS spatial surfaces 2018, Meta Relative Wealth Index, travel time to healthcare, ACLED, poverty.

Method 2: zero-dose rate and its uncertainty, all 774 LGAs

Every LGA has its own estimate and its own 95% credible interval.





Model estimate. Method 2 (Bayesian small-area estimation), 2026. n = 774 LGAs.

30.6%

national zero-dose rate, 2026 (95% CrI 26.5 to 35.0%)

50.6%

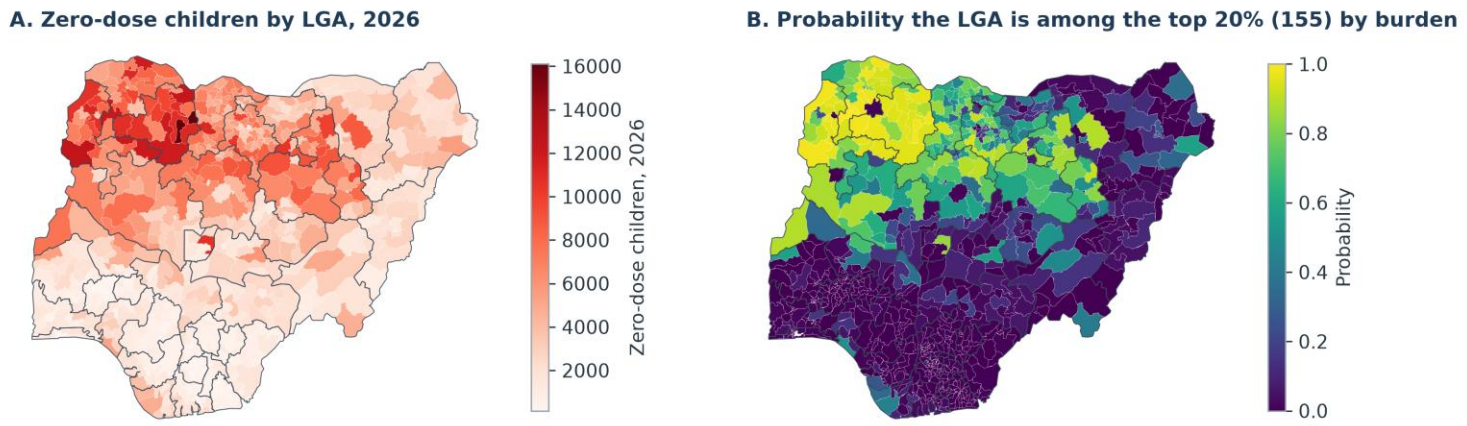
North-West zone rate, the highest of the six zones

How to read the right-hand map. Wider intervals mark LGAs where survey anchors and local measures leave more room for error. These LGAs are the first candidates for local verification, for example a rapid coverage assessment, before large investments.

Model estimate. Method 2: Bayesian small-area estimation (SAE), 2026. CrI = credible interval. Sources: NDHS 2008-2024; NPC 2022; GRID3 boundaries; covariates as listed on the method slide.

Method 2: zero-dose burden and priority probability

Burden combines rate and child numbers; the probability shows how stable each priority is.



Model estimate. Method 2, 2026. n = 774 LGAs.

2.15 million

zero-dose children in 2026 (95% CrI 1.86 to 2.46 million)

54%

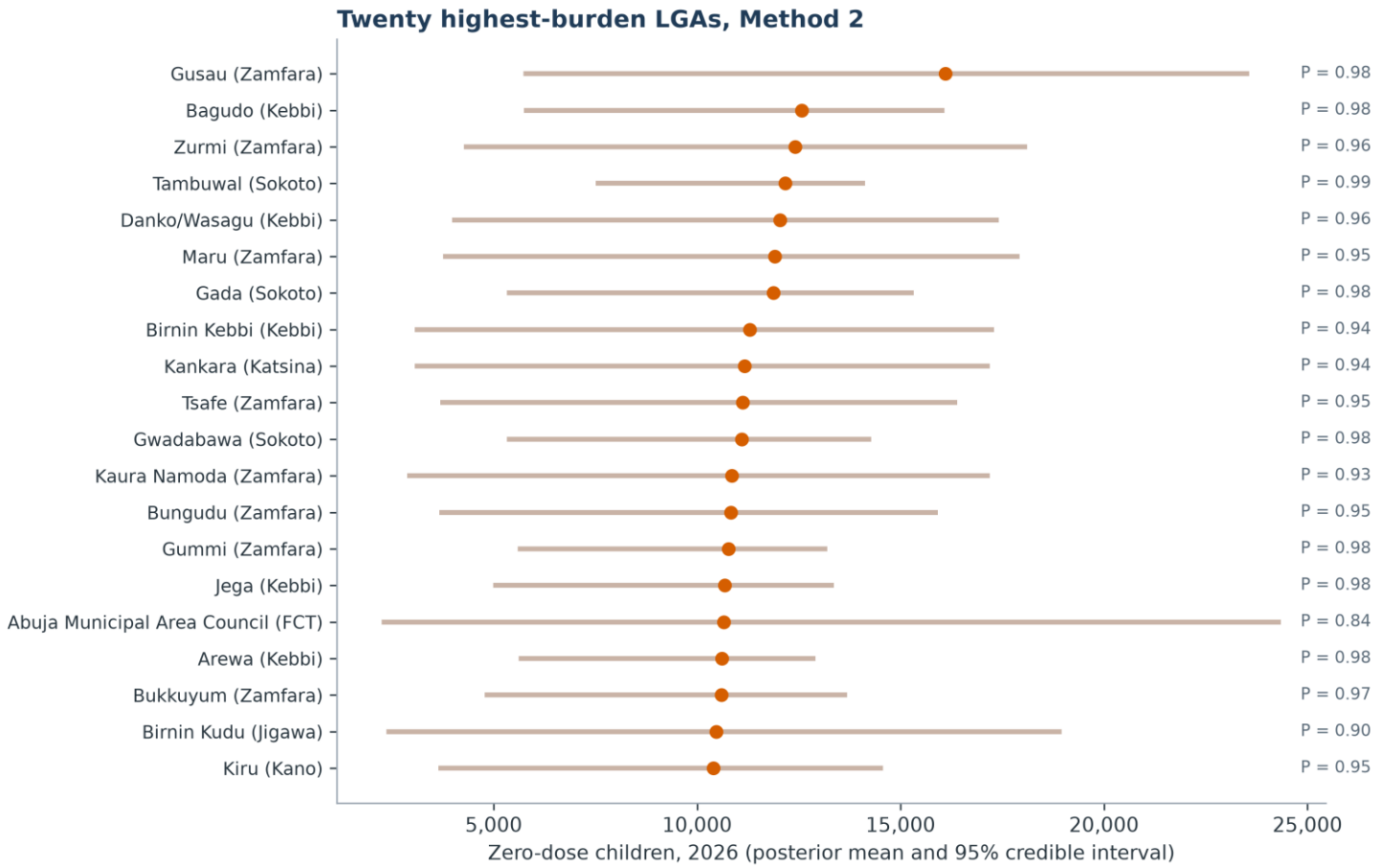
of the burden is in the top 20% of LGAs (155 LGAs)

Priority probability. For each LGA, the share of 4,000 model simulations in which it ranks among the 155 highest-burden LGAs. 40 LGAs are priorities in at least 90% of simulations; LGAs between 0.1 and 0.9 are borderline and should be confirmed with local data.

Model estimate. Method 2: Bayesian small-area estimation (SAE), 2026. Children aged 12-23 months from the 2024 under-five projection divided by five, shared by NPC 2022 LGA population.

Method 2: the 20 highest-burden LGAs

Each bar shows the 95% credible interval; P is the probability of being among the top 155.



Model estimate. Method 2, 2026.

19 of 20

are in the North-West zone

19 of 20

are priorities in at least 90% of simulations

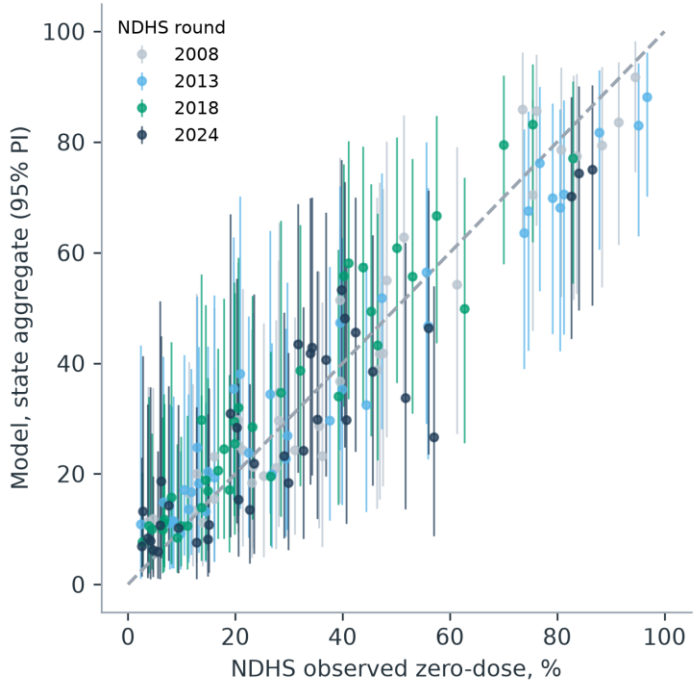
Concentration. Zamfara 8, Kebbi 5, Sokoto 3, Katsina 1, FCT 1, Jigawa 1, Kano 1. These LGAs combine high rates with large child populations.

Model estimate. Method 2: Bayesian small-area estimation (SAE), 2026. Ranks from 4,000 posterior draws.

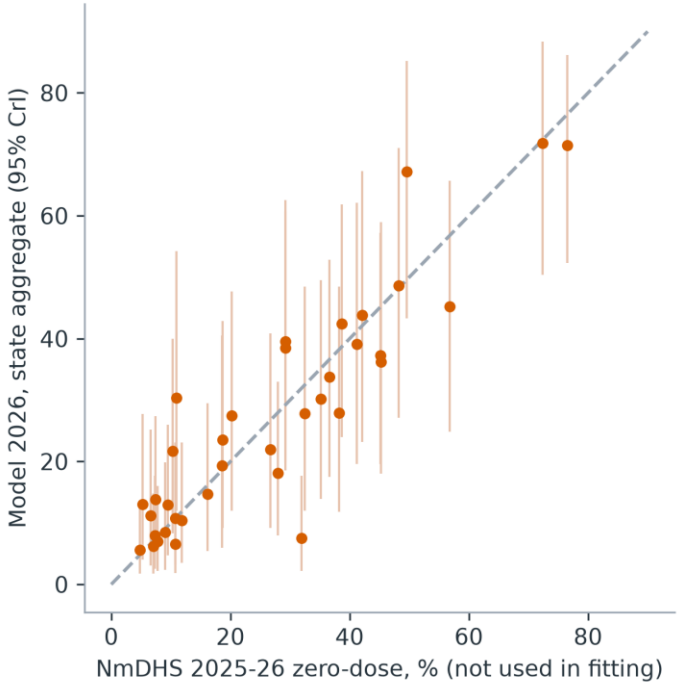
Method 2: model performance and diagnostics

The model fits every survey round, passes an independent survey check, and its sampler converged cleanly.

A. Fit to 148 survey results: 98% in 95% PI



B. Independent survey: rho = 0.87, MAE = 5.9 pp



C. Convergence and calibration

Diagnostic	Value
Chains x draws	4 x 8,000
Parameters monitored	1,649
Maximum R-hat	1.003
Minimum bulk ESS	2,549
Minimum tail ESS	2,524
Divergent transitions	0
Survey 95% PI coverage	98%
Neighbour pairs (BYM2)	2,165

PI = posterior predictive interval including survey sampling error. R-hat <= 1.01 and no divergences indicate convergence.

1.003
maximum R-hat (target below 1.01)

2,549
minimum effective sample size (target above 400)

0
divergent transitions

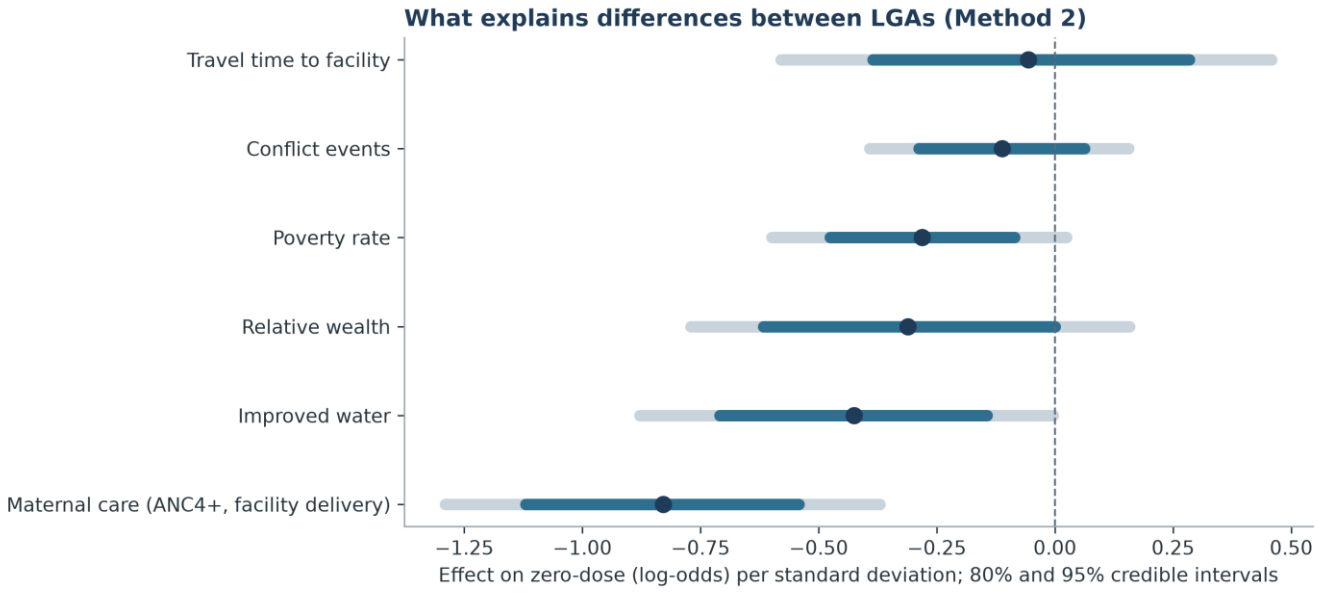
98%
of NDHS state-round values inside the 95% predictive interval

rho 0.87
against NmDHS 2025-26 (37 states); mean absolute error 5.9 points

Method 2: Bayesian small-area estimation (SAE). 1,649 parameters; 4 chains x 8,000 draws after 3,000 tuning steps. NmDHS = Nigeria mini Demographic and Health Survey, not used in fitting.

Method 2: what explains differences between LGAs

Maternal care is the strongest and most certain predictor of zero-dose within states.



Model estimate. Negative = lower zero-dose. Covariates: DHS spatial surfaces 2018, Meta RWI, Weiss travel time, ACLED, poverty surface.

Local measure	Effect (95% CrI)	Odds ratio
Maternal care	-0.83 (-1.29 to -0.37)	0.44
Improved water	-0.42 (-0.88 to -0.00)	0.65
Relative wealth	-0.31 (-0.77 to +0.16)	0.73
Travel time to facility	-0.06 (-0.58 to +0.46)	0.95
Conflict events	-0.11 (-0.39 to +0.16)	0.89
Poverty rate	-0.28 (-0.60 to +0.02)	0.75

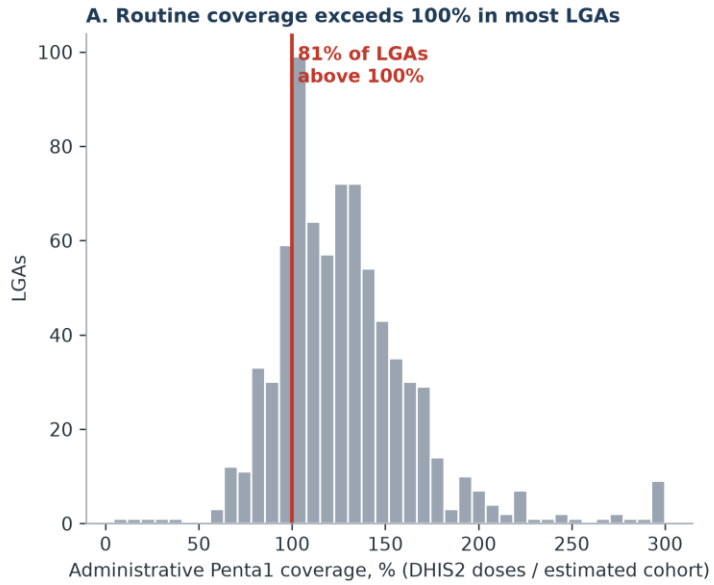
Associations, not causes. These are partial associations used to distribute survey levels within states, not causal effects.

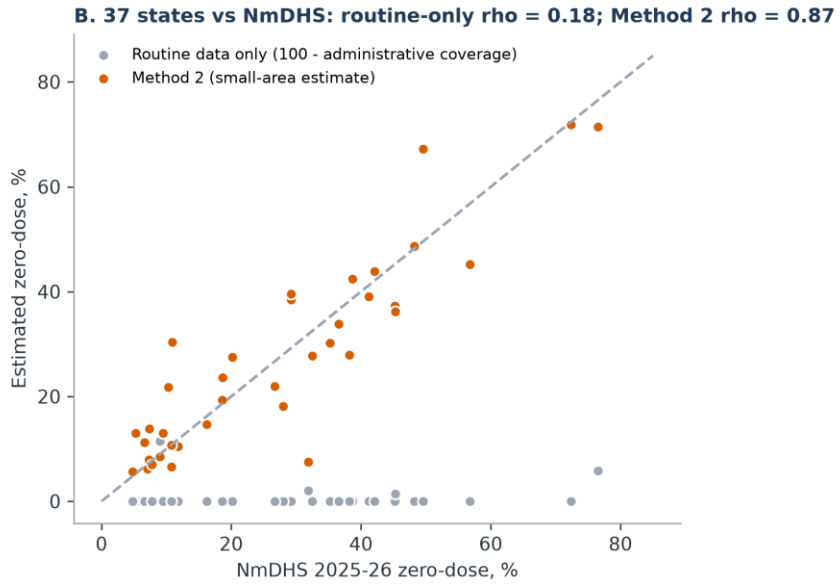
Reading the table. Effects are on the log-odds scale per standard deviation; below zero means lower zero-dose. An odds ratio of 0.44 means one standard deviation better maternal care is linked with about 56% lower odds of a child being zero-dose.

Model estimate. Method 2: Bayesian small-area estimation (SAE). Maternal care = mean of standardised ANC4+ and facility delivery (DHS spatial surfaces, 2018).

Routine coverage and survey-anchored estimates

Why Method 2 uses survey levels rather than routine coverage to estimate LGA zero-dose.





DHIS2 Penta1 2023-2025 annual mean over the 2024 cohort projection (state under-5 / 5, NPC 2022 shares). Sources: DHIS2; NmDHS 2025-26.

81%

of LGAs show Penta1 coverage above 100%

rho 0.18

routine-only zero-dose vs NmDHS 2025-26, 37 states

rho 0.87

Method 2 vs NmDHS 2025-26, 37 states

What this means. Administrative coverage divides doses given by an estimated target population. When the target population is too small, or doses include children from other LGAs, coverage exceeds 100 percent and cannot be read as a rate. Routine data remain essential for monitoring trends and data quality; for the level of zero-dose in each LGA, survey-anchored estimates are more reliable (routine-only state estimates differ from NmDHS by 27 points on average).

DHIS2 Penta1 2023-2025 annual mean over the 2024 cohort projection (state under-five / 5, NPC 2022 shares). Sources: DHIS2; NmDHS 2025-26.

Method 1 and Method 2 compared by zone

About half of zero-dose children are in the North-West under both methods; national totals differ by 2.4%.

Zone	LGAs (M1 / M2)	Children 12-23 months	Method 1 zero-dose children (95% interval)	Method 1 rate %	Method 2 zero-dose children (95% CrI)	Method 2 rate %	Difference M2 vs M1	NmDHS 2025-26 %
North West	186 / 186	2,269,638	1,084,155 (961,608 - 1,201,516)	47.8	1,149,017 (923,311 - 1,371,914)	50.6	+6.0%	48.7
North East	111 / 112	1,073,384	336,687 (255,921 - 417,822)	31.4	372,892 (265,189 - 495,882)	34.7	+10.8%	38.7
North Central	121 / 121	1,039,371	343,731 (262,802 - 433,604)	33.1	324,138 (228,447 - 421,964)	31.2	-5.7%	34.2
South West	137 / 137	1,068,159	154,451 (97,752 - 231,362)	14.5	141,155 (86,336 - 210,104)	13.2	-8.6%	14.4
South South	123 / 123	821,850	112,309 (67,941 - 169,284)	13.7	110,354 (65,945 - 169,111)	13.4	-1.7%	8.8
South East	95 / 95	746,305	67,871 (36,662 - 111,588)	9.1	52,005 (26,981 - 87,430)	7.0	-23.4%	11.2
Nigeria	**773 / 774**	**7,018,707**	**2,099,204** (1,898,902 - 2,308,432)	**29.9**	**2,149,561** (1,856,459 - 2,455,411)	**30.6**	**+2.4%**	**-**

Same national picture. 2.10 million (Method 1) and 2.15 million (Method 2); the 95% intervals overlap in every zone.

North-West first. 52% of zero-dose children under Method 1 and 53% under Method 2.

Largest gap: North-East. Method 2 is 11% higher (34.7% vs 31.4%); NmDHS 2025-26 shows 38.7%.

Model estimate, 2026. Method 1: Bayesian hierarchical model with DHIS2-calibrated LGA allocation (773 LGAs); Method 2: Bayesian small-area estimation (SAE) (774 LGAs). NmDHS 2025-26 zone rates for reference (not used in fitting).

Method 1 and Method 2 compared by state

The methods rank the 37 states almost identically (rho = 0.96); ordered by Method 1 burden.

State	Method 1 children	Method 2 children	M1 rate %	M2 rate %	NmDHS %
Kano	198,101	227,157	36.8	42.4	38.7
Katsina	169,685	211,744	39.0	48.6	48.2
Sokoto	166,648	167,720	72.7	71.4	76.5
Kebbi	153,659	162,123	69.0	71.8	72.4
Zamfara	150,220	143,697	70.9	67.2	49.6
Kaduna	139,578	119,060	42.0	37.3	45.2
Bauchi	111,948	118,781	33.3	36.2	45.3
Jigawa	106,264	117,518	34.8	38.5	29.2
Niger	104,030	104,633	45.0	45.2	56.8
Borno	60,604	70,746	30.3	39.1	41.2
Benue	52,312	42,277	28.9	23.6	18.7
Gombe	50,128	57,947	34.2	39.6	29.2
Kogi	48,476	41,779	31.3	27.5	20.2
Kwara	47,518	42,528	36.6	33.8	36.6
Oyo	45,265	42,253	19.3	18.1	28.0
Yobe	43,282	53,059	35.3	43.8	42.1
Plateau	42,309	44,941	27.9	30.2	35.2
Taraba	35,731	34,085	29.2	27.9	38.2
Adamawa	34,994	38,275	20.1	22.0	26.7

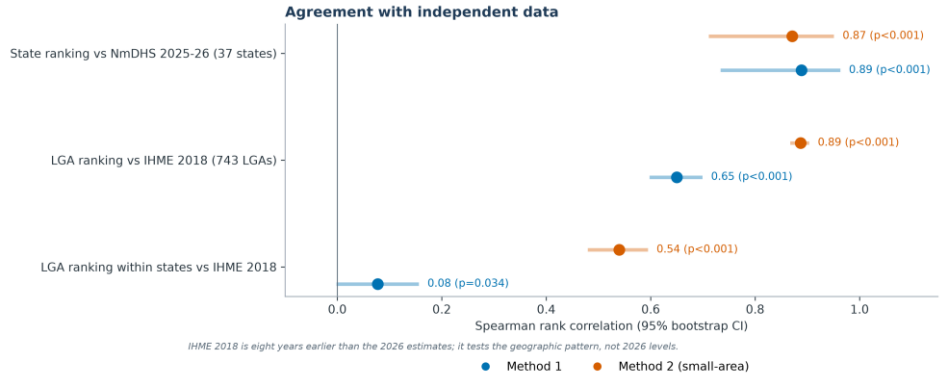
State	Method 1 children	Method 2 children	M1 rate %	M2 rate %	NmDHS %
Ogun	32,123	22,819	18.9	14.7	16.2
Rivers	30,795	26,122	14.7	13.0	9.5
Nasarawa	28,735	29,256	25.6	27.8	32.5
Ondo	28,122	32,119	18.7	21.7	10.3
Lagos	22,095	18,709	7.1	6.2	7.1
Delta	21,940	23,510	12.9	13.9	7.4
FCT	20,351	18,725	18.5	19.3	18.6
Akwa Ibom	19,144	13,242	13.1	10.5	11.8
Anambra	18,507	14,459	9.8	7.9	7.3
Edo	15,905	12,516	10.0	8.5	9.0
Enugu	14,715	10,442	9.9	7.0	7.8
Osun	14,337	12,763	12.1	10.7	10.8
Imo	14,271	9,100	8.5	5.6	4.8
Abia	12,741	10,508	8.9	7.5	31.9
Ekiti	12,509	12,492	11.2	11.2	6.6
Bayelsa	12,434	20,938	18.0	30.3	10.9
Cross River	12,091	14,026	11.2	13.0	5.3
Ebonyi	7,637	7,496	6.7	6.6	10.8

Model estimate, 2026. Method 1: Bayesian hierarchical model with DHIS2-calibrated LGA allocation; Method 2: Bayesian small-area estimation (SAE). Children = modelled zero-dose children aged 12-23 months. NmDHS 2025-26 for reference.

National comparison and agreement with independent data

Both methods match the new national survey; Method 2 agrees more closely with the independent IHME LGA map.

Measure	Method 1	Method 2 (SAE)
LGAs estimated	773	774
Zero-dose children, 2026 (95% interval)	2.10 million (1.90 - 2.31)	2.15 million (1.86 - 2.46)
National zero-dose rate	29.9%	30.6%
NmDHS 2025-26, 37 states: Spearman rho (95% CI)	0.89 (0.74 to 0.96)	0.87 (0.71 to 0.95)
NmDHS 2025-26: mean absolute error / bias, points	6.3 / -0.7	5.9 / +0.1
IHME DTP1 2018, 743 LGAs: Spearman rho (95% CI)	0.65 (0.60 to 0.70)	0.89 (0.87 to 0.90)
IHME within states: Spearman rho (95% CI)	0.08 (0.00 to 0.15)	0.54 (0.48 to 0.59)
States with positive within-state agreement	22 of 37	36 of 37

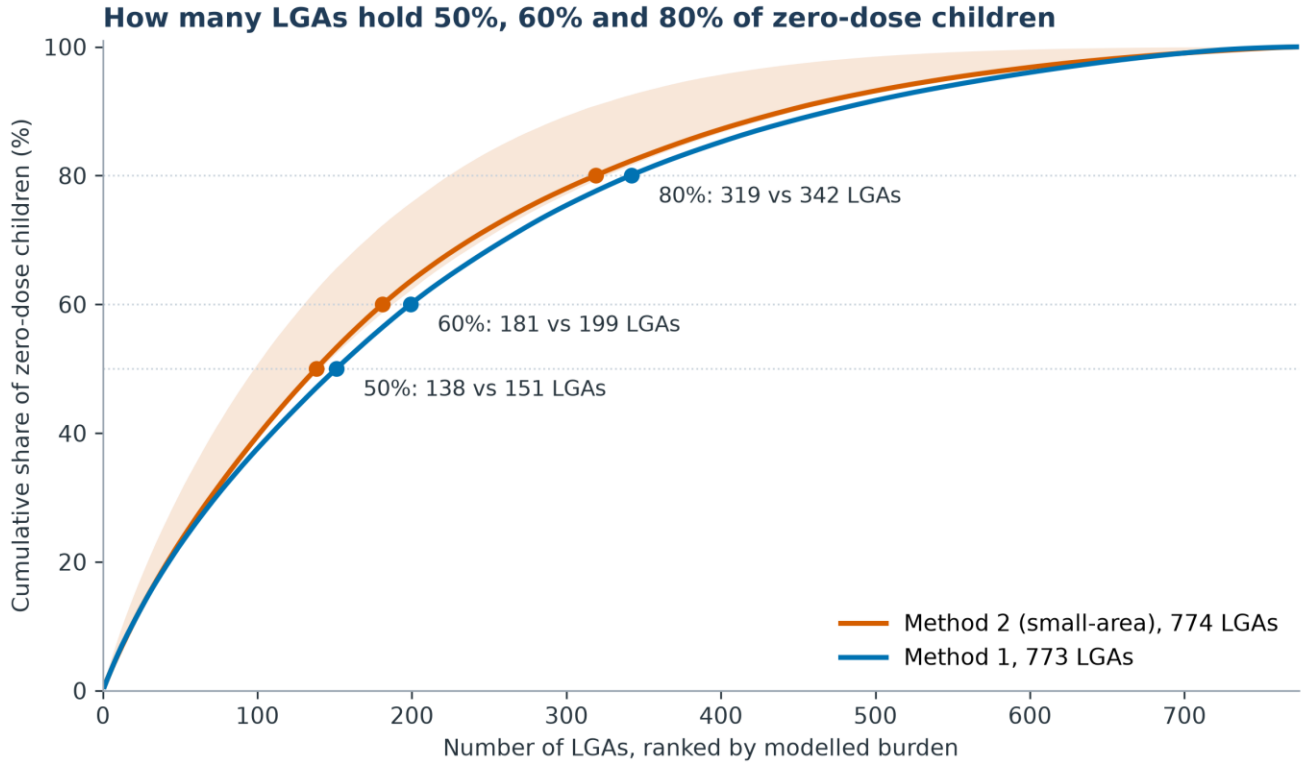


How to read this. Both methods rank states almost identically against NmDHS 2025-26, which neither model used. At LGA level, Method 2 agrees far more closely with the independent IHME map, above all within states, where Method 1 follows routine reporting patterns. 107 of the top 155 LGAs are shared by both methods.

rho = Spearman rank correlation; 95% CI from 2,000 bootstrap resamples. All p < 0.001 except Method 1 within states (p = 0.03). IHME = Institute for Health Metrics and Evaluation Local Burden of Disease DTP1, 2018 (zero-dose = 100 - DTP1).

LGAs needed to reach 50, 60 and 80 percent of the burden

Burden concentration under both methods, ranking LGAs from highest to lowest modelled burden.



Model estimate, 2026.

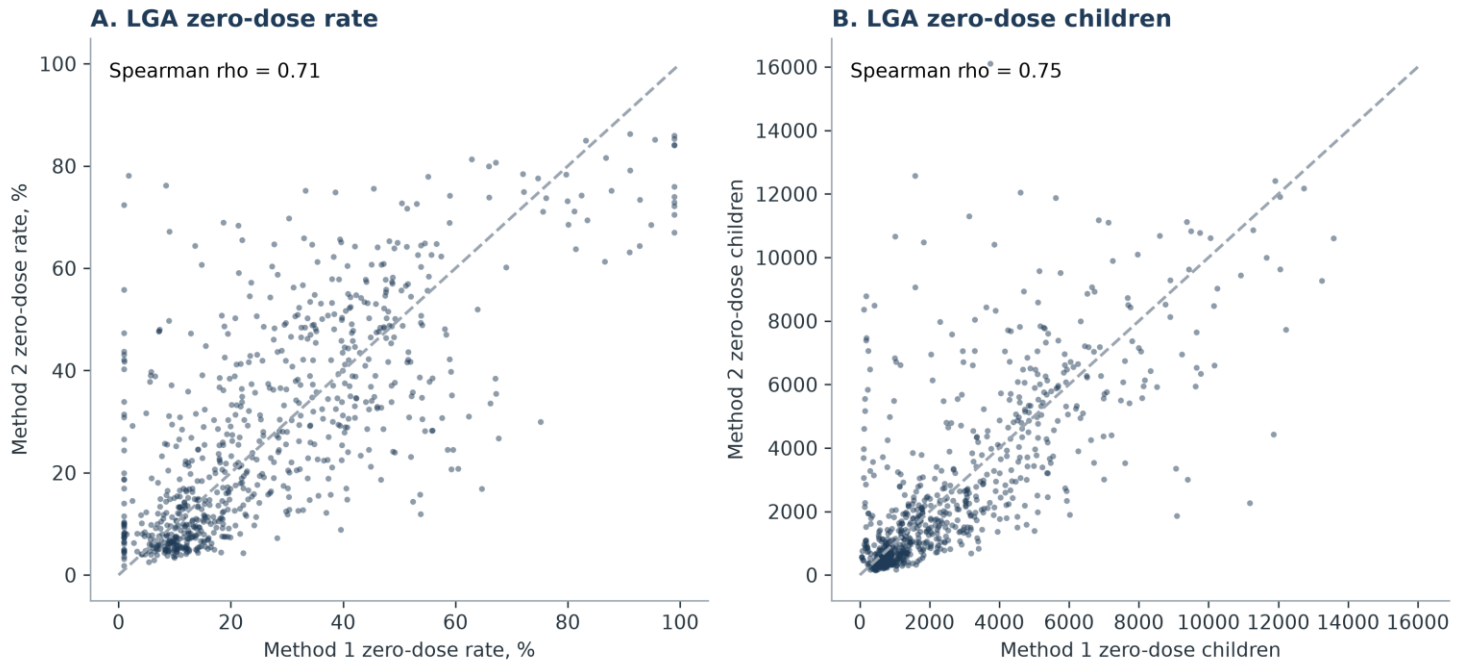
Share of zero-dose children	Method 1 LGAs (% of LGAs)	Method 2 LGAs (% of LGAs)
50%	**151 (20%)**	**138 (18%)**
60%	**199 (26%)**	**181 (23%)**
80%	**342 (44%)**	**319 (41%)**
Held by top 155 (20% of LGAs)	50.9% (46.9 - 54.7)	54.2% (49.6 - 58.6)

Planning rule of thumb. About 140 to 150 LGAs reach half of all zero-dose children, about 180 to 200 reach 60 percent, and 320 to 340 reach 80 percent, whichever method is used. Method 2 is slightly more concentrated.

Model estimate, 2026. Method 1: Bayesian hierarchical model with DHIS2-calibrated LGA allocation (773 LGAs); Method 2: Bayesian small-area estimation (SAE) (774 LGAs). Method 2 band = 95% credible interval of the cumulative share.

Where the two methods agree and differ at LGA level

Strong agreement overall; differences where routine reporting is out of line with population.



Model estimate, 2026.

rho = 0.75

rank agreement in zero-dose children across 773 LGAs

107 of 155

top-20% LGAs shared by both methods: the most robust priorities

Where they differ. Method 1 follows each LGA's share of reported Penta1 doses: very high reported doses push its rate toward the 1% floor (for example Rabah and Goronyo, Sokoto) and low reported doses push it up (for example Sokoto North). Method 2 is not affected by reporting volumes. LGAs with large gaps are priorities for data-quality review.

Model estimate, 2026. Method 1: Bayesian hierarchical model with DHIS2-calibrated LGA allocation; Method 2: Bayesian small-area estimation (SAE). Spearman rank correlation of modelled zero-dose children.

Which estimate for which decision

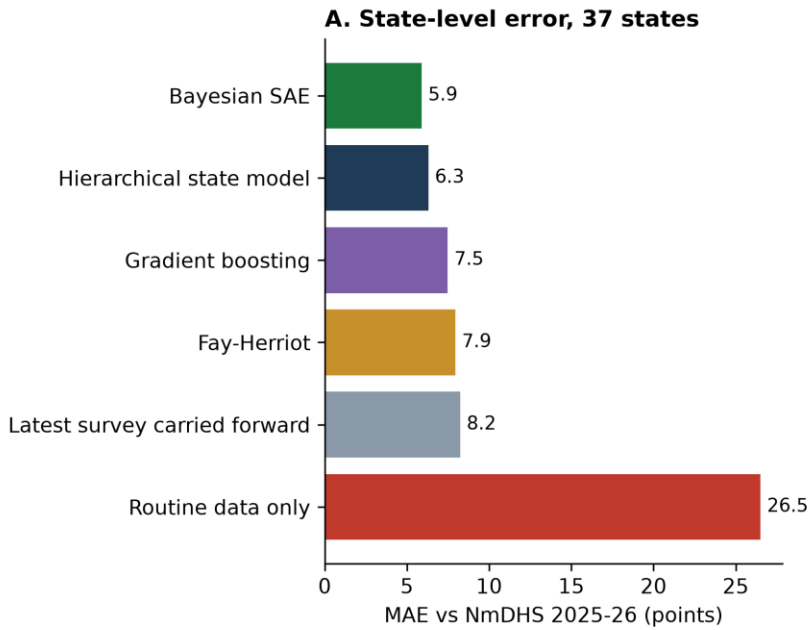
The two methods answer the same question in different ways; each has a role.

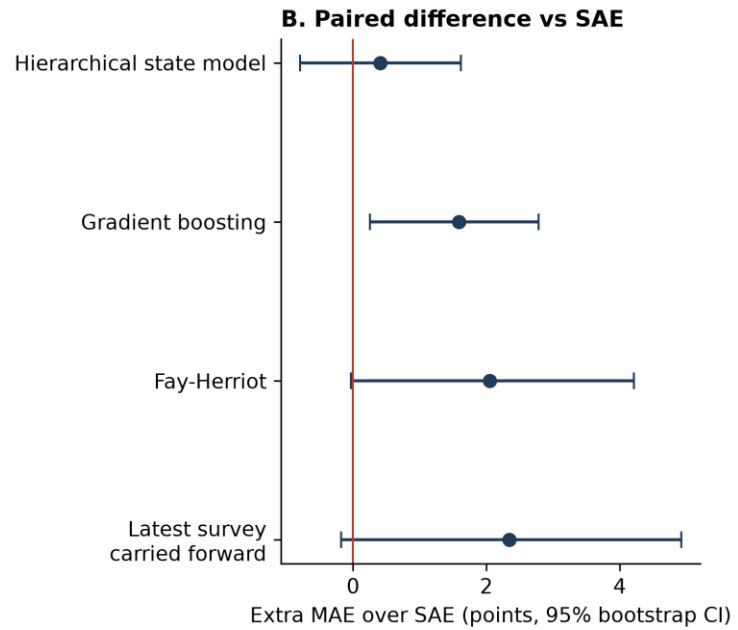
Decision	Recommended estimate	Why
National, zone and state totals	**Either; report Method 1 with Method 2 as the sensitivity range**	Totals agree within 2.4% nationally; state rank agreement rho = 0.96; both match NmDHS 2025-26
Ranking LGAs within a state	**Method 2 (SAE)**	Covers all 774 LGAs; full uncertainty for each LGA; within-state agreement with IHME rho = 0.54 vs 0.08
Selecting priority LGAs for funding	**The 107 LGAs in the top 155 under both methods first**	Robust to the choice of method; then add LGAs with high Method 2 priority probability
Tracking change and early warning	**Method 1**	Uses the DHIS2 trend and updates with monthly routine data
Data-quality review	**LGAs where the methods differ most**	Large gaps flag reported doses out of line with population

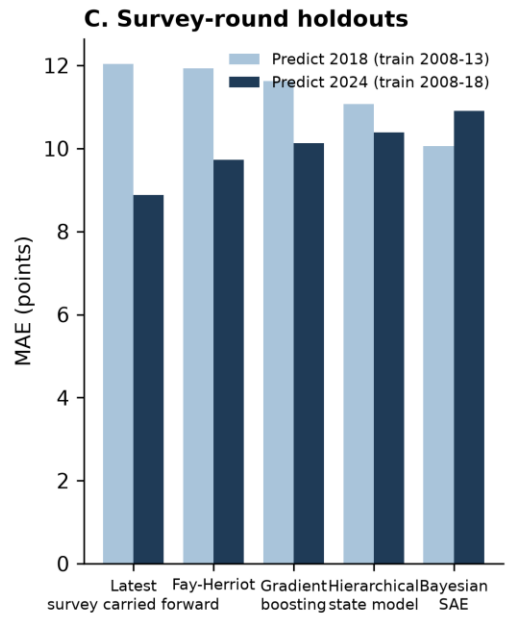
Method 1: Bayesian hierarchical model with DHIS2-calibrated LGA allocation; Method 2: Bayesian small-area estimation (SAE). IHME = Institute for Health Metrics and Evaluation; NmDHS = Nigeria mini Demographic and Health Survey.

Method 2 is the most accurate approach tested

Independent state-level check against the NmDHS 2025-26 (37 states); the survey was not used to fit any model.







Routine data MAE 26.5 shown in A. Differences in B include 0 for all comparators except gradient boosting. Source: A02 tables.

5.9 points

mean absolute error (MAE), lowest of six approaches; bias +0.1

0.87

rank correlation with the survey (95% CI 0.73 to 0.95)

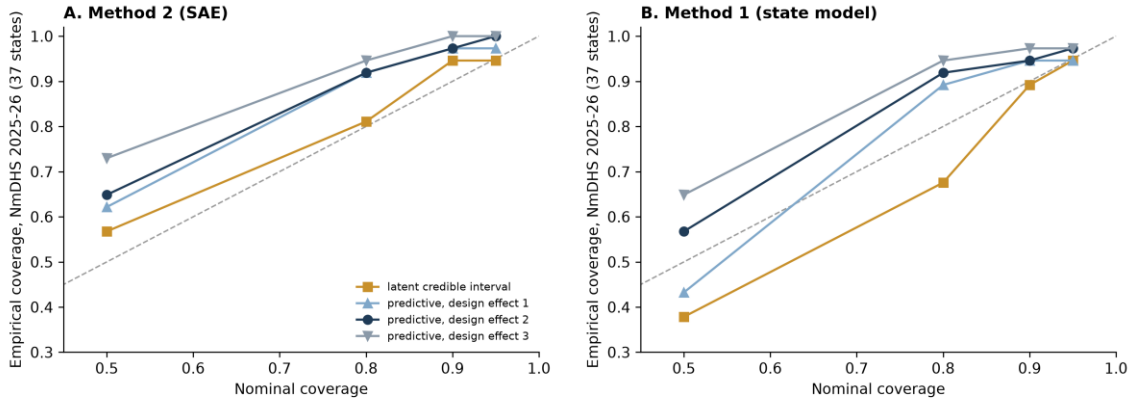
+0.4 points

Method 1 difference (95% CI -0.8 to 1.6): statistically equivalent

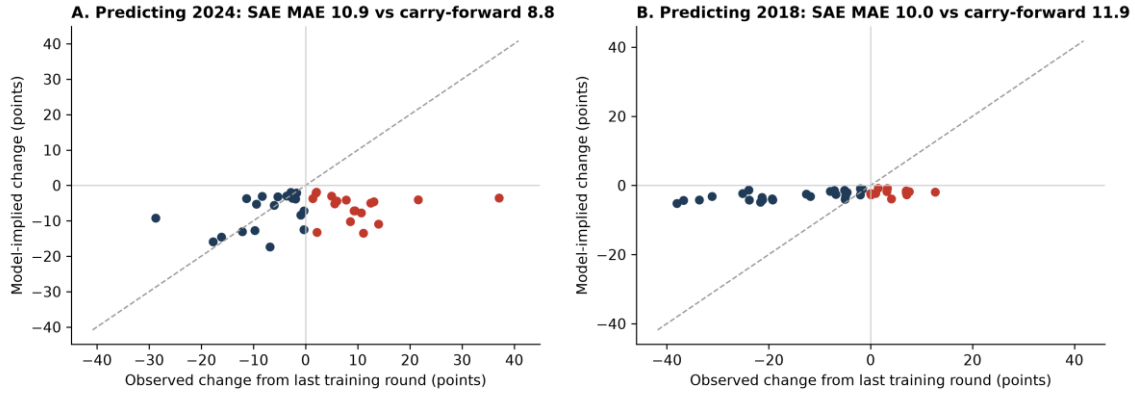
Model estimate. MAE in percentage points; CI = bootstrap confidence interval over states. Comparators: Method 1 (6.3), gradient boosting (7.5), Fay-Herriot area-level model (7.9), latest survey carried forward (8.2), routine data only (26.5).

Honest uncertainty; forecasting ahead is the weak link

Intervals cover the survey as intended; projecting trends forward is harder than describing the present.



Above the dashed line = intervals wider than needed. With 37 states, one state = 2.7 points of coverage. Source: A_R2.



Red = model projected the wrong direction of change. 2018-2024 stagnation broke the 2008-2018 decline. Source: A_R4.

Calibrated. With design effect 2, the 95% predictive interval contained the NmDHS value in 37 of 37 states and the 50% interval in 24 of 37, as expected (p = 0.10 and 0.26). Continuous ranked probability score (CRPS) 4.4, against 4.5 for Method 1.

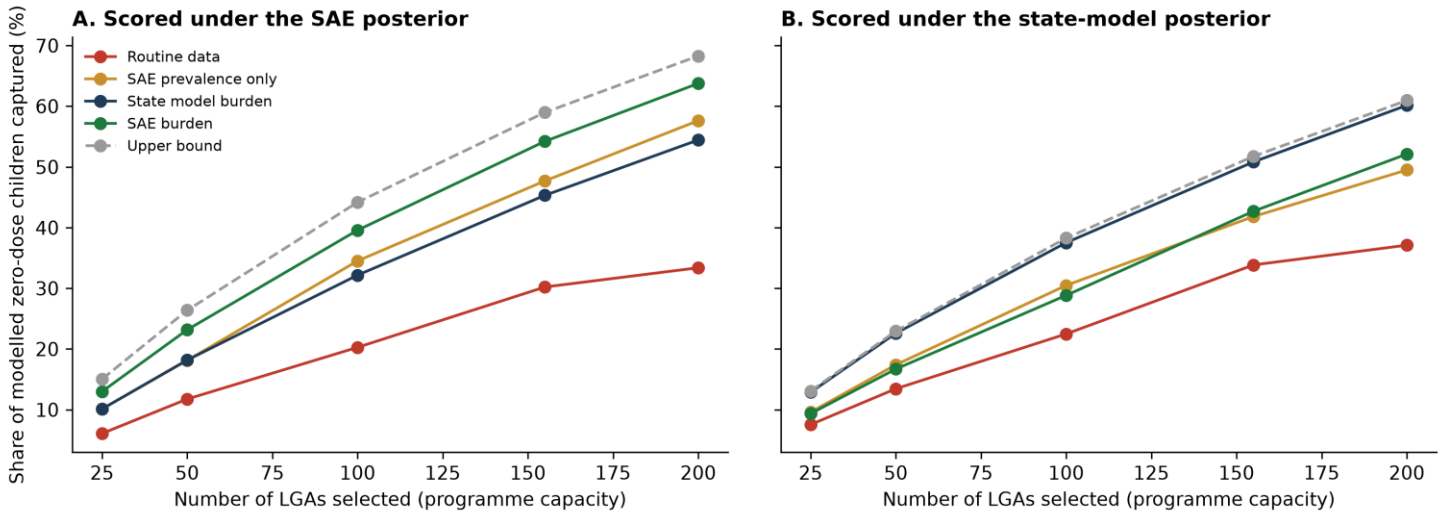
Trend breaks. With survey rounds held out, Method 2 beat carry-forward for 2018 (MAE 10.1 vs 12.0) but not for 2024 (10.9 vs 8.9). In 18 states the projected direction was wrong, with errors nearly twice as large (14.3 vs 7.7).

What it means. Use the intervals, not only the point estimates, and refresh the model with every new survey round: the 2026 estimates carry real trend uncertainty.

Model estimate. Design effect 2 allows for survey sampling error. MAE = mean absolute error, percentage points.

Prioritizing under uncertainty

Fund near-certain priorities first; verify uncertain LGAs through microplanning.



Model-based capture of modelled children (no LGA ground truth). Each model favours its own list; routine data is lowest under both. Source: A_R3.

Capture. Ranking by rate alone captures 48%, and ranking by routine administrative coverage 30-34%. Each model favours its own list, so these are model-based, not observed.

Robust. State MAE stays at 5.86-6.02 across design-effect, prior and spatial choices; the 2025 population projection raises the total by 12.7% but changes only 4 of the 155 priority LGAs.

40 LGAs

near-certain to be in the top 155 (probability 0.9 or more)

217 LGAs

uncertain; 102 of them fall outside the ranked list of 155

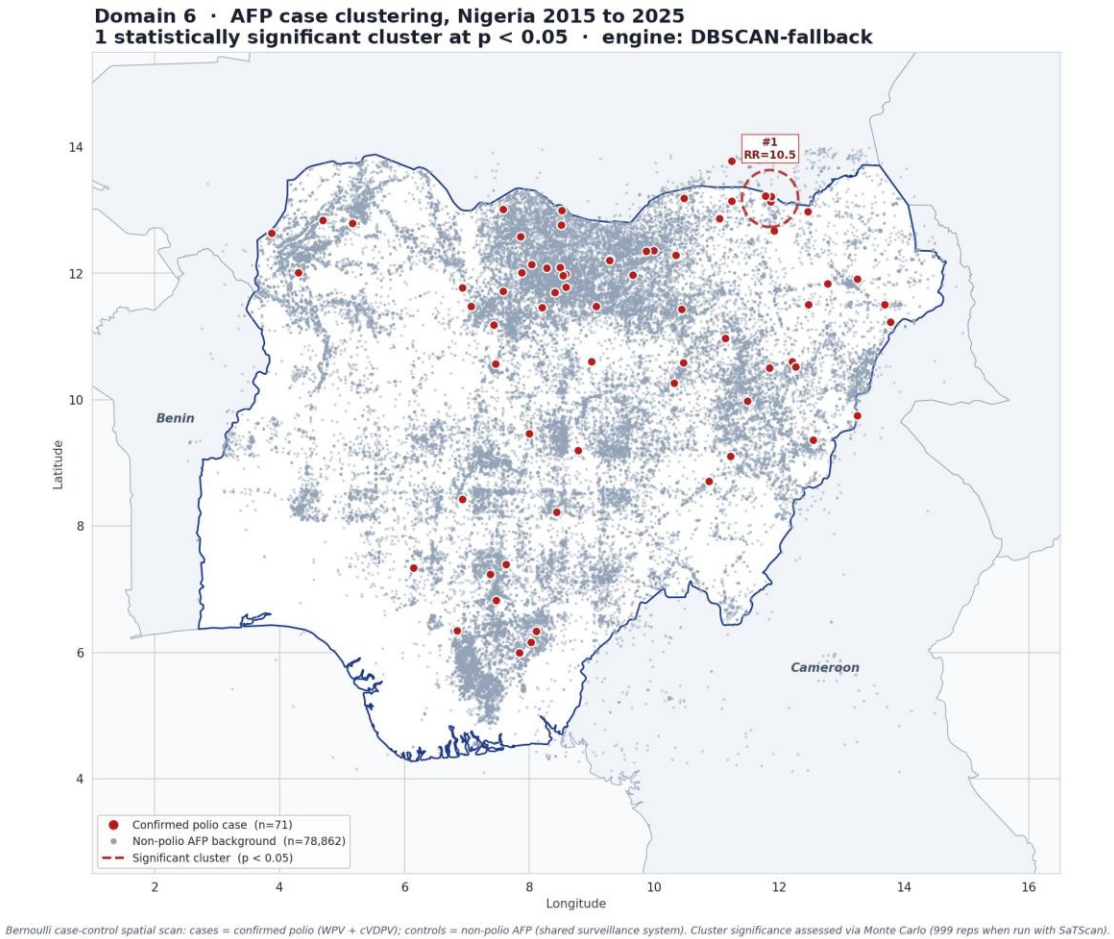
54%

of modelled zero-dose children in the 155 LGAs ranked by Method 2 expected burden

Model estimate. Method 2: Bayesian small-area estimation (SAE), 4,000 posterior draws. MAE = mean absolute error.

Where is polio-detection risk highest?

One cluster on the north-east border, and six states that have not yet reached the three-year mark.



RR = 10.5

the highlighted border area needs closer surveillance and response

Meaning: the highlighted area should remain a priority for surveillance, investigation and response.

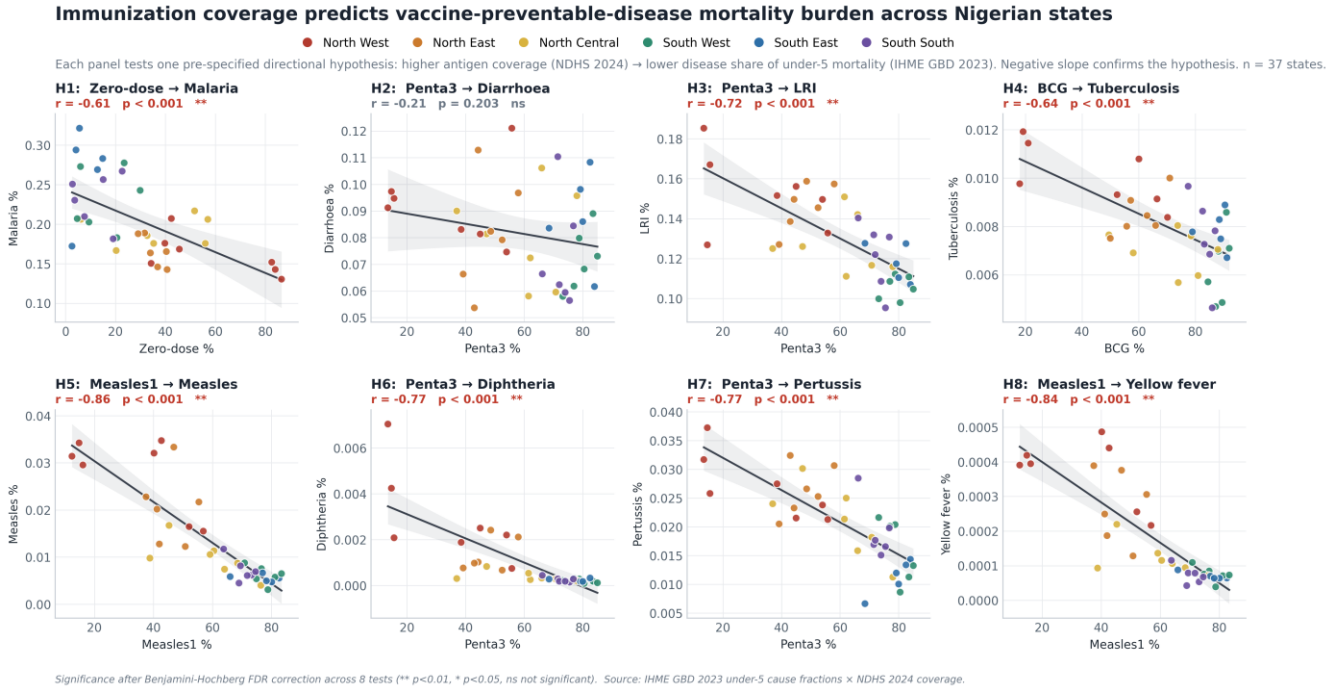
Programme meaning: Taraba, Kogi, Adamawa, Borno, Gombe and Kano require continued attention.

Action: maintain strong campaign quality, AFP surveillance, environmental surveillance and timely investigation.

Model estimate. Source: Domain 6 spatial scan on the national acute flaccid paralysis line list (about 103,000 records, 2015 to 2025); survival analysis against the WHO three-year certification rule.

Domain 7: outbreak risk and the link to child deaths

Places with lower vaccination coverage also show higher risk of child deaths from vaccine-preventable diseases.



7 of 8

pre-specified coverage-to-mortality links hold after correction for multiple testing

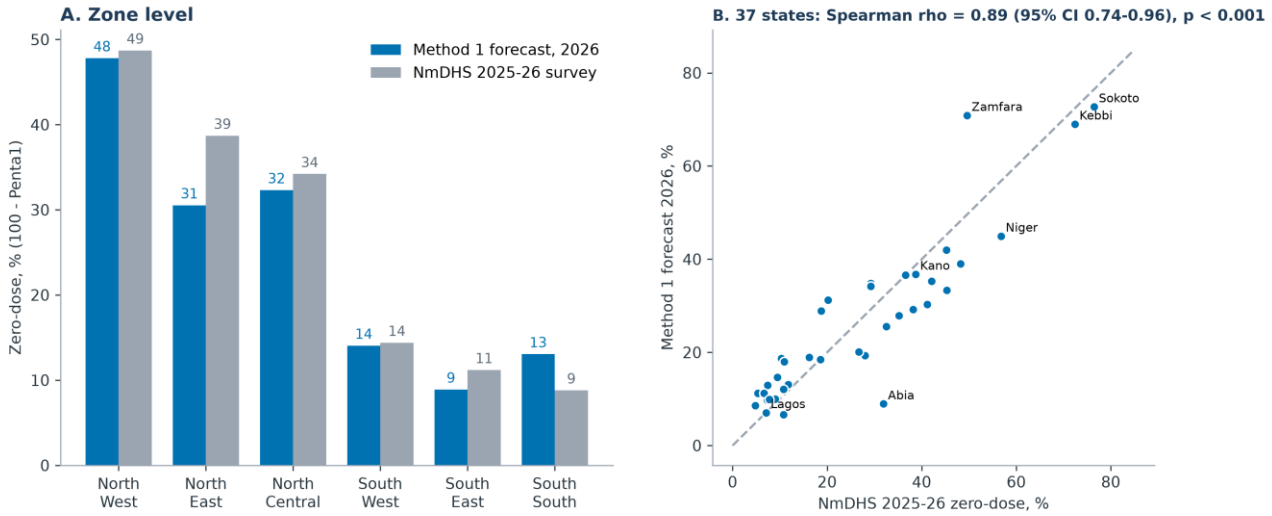
Strongest for measles. The strongest link was for measles. Improving measles coverage in high-risk areas can protect more children.

Why it matters: reaching missed children is not just about percentages; it is about preventing illness and death.

Model estimate. Source: Domain 7, eight pre-specified false-discovery-rate-corrected tests against IHME Global Burden of Disease 2023 cause fractions, a validated proxy while the national surveillance line-lists are pending.

External check, state level

Model estimate against out-of-sample data. The NmDHS 2025-26 survey was released after this work concluded.



Model estimate vs survey. MAE 6.3 percentage points. Sources: NmDHS 2025-26 Table 10 (100 - Penta1); Method 1 forecast.

rho = 0.89

Our forecast and the new survey rank the 37 states in almost the same order.

48% vs 49%

our North-West forecast against the new survey - within one point

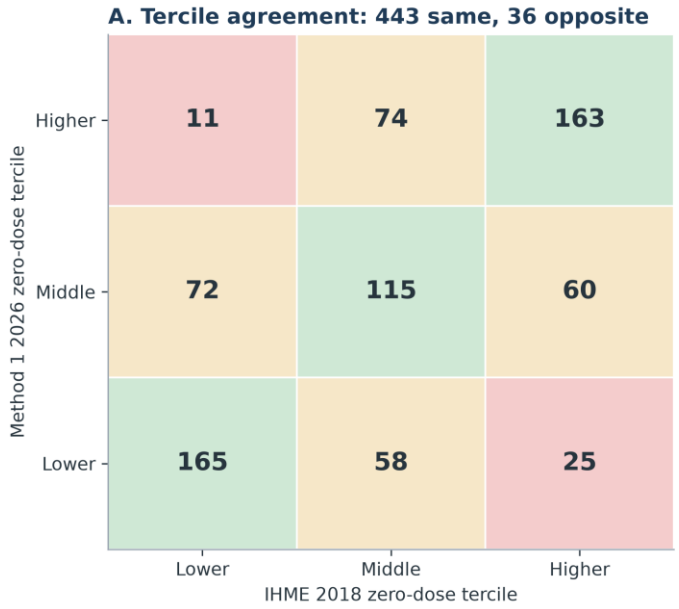
Why this is the strongest evidence. The NmDHS 2025-26 did not exist when the model was built - a true out-of-sample test.

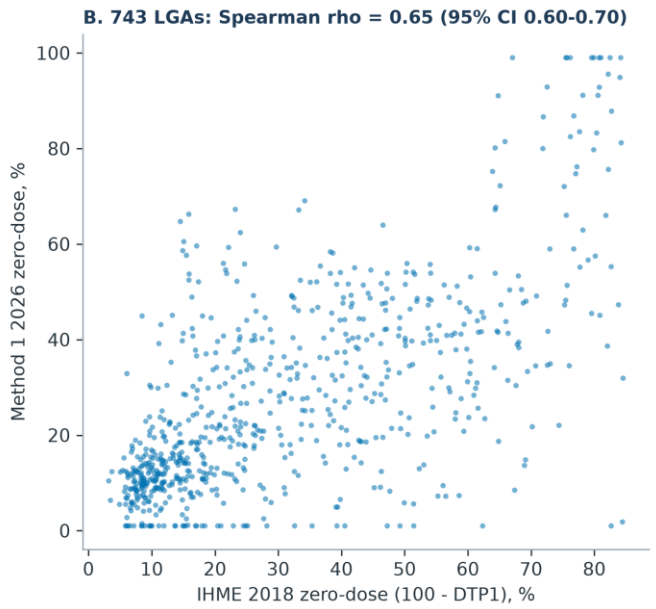
The pattern also matched independent child-death data.

rho = Spearman rank correlation, where 1.00 is a perfect match of the ordering. Sources: Nigeria mini Demographic and Health Survey (NmDHS) 2025-26; consortium forecast; IHME Global Burden of Disease 2023. Out-of-sample check.

External check, local-government level

Independent data, built another way, puts the same areas in the same order.





Cross-year comparison (2026 model vs 2018 IHME). Sources: IHME LBD DTP1 admin-2 2000-2018; Method 1.

rho = 0.65

Our LGA ranking and the independent map show similar priority patterns.

Most LGAs were placed in the same or a neighbouring priority third. Only 36 were in opposite thirds and should be reviewed.

Why the levels differ. The IHME surface ends in 2018 and ours is for 2026. Eight years shift the levels, but the geography still agrees.

Model estimate: Method 1 local-government estimate as DTP1 coverage (100 minus zero-dose), against the IHME Local Burden of Disease DTP1 admin-2 surface, 2000 to 2018. rho = Spearman rank correlation, 743 matched LGAs.

How the RI team uses this

Each output maps to a decision the programme already makes.

1 Microplanning

Use the priority list and vaccine-risk calendar in annual RI workplans.

2 Barrier-matched packages

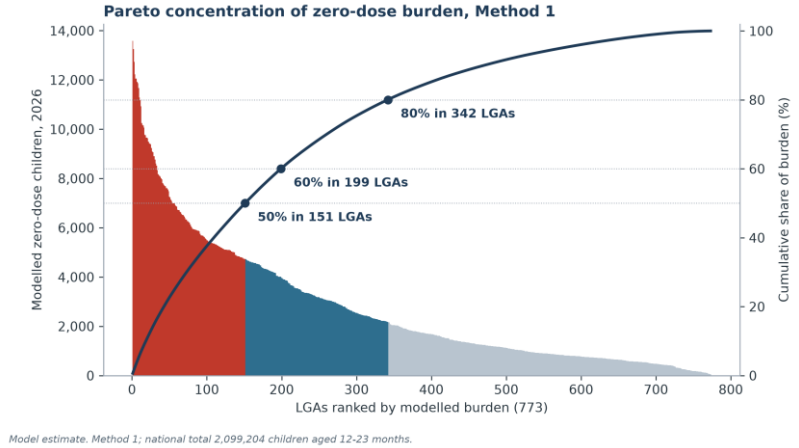
Use each LGA's measured barriers to choose the package, starting from the state package.

3 Campaign and supervision routing

Use hotspot and burden ranking to choose campaign areas and supervision routes.

4 Leverage beyond immunization

Many high-risk places also need nutrition and antenatal-care support; one visit can deliver several services.



The tool ranks LGAs by estimated missed children and shows how much of the burden each group contains.

Sequencing the money. Start with about 151 LGAs to reach half of missed children, expand to about 199 for 60 percent, then toward 342 LGAs to reach 80 percent.

Model estimate. Source: Domain 5 local-government ranking and Pareto curve (Method 1: Bayesian hierarchical model with DHIS2-calibrated LGA allocation; 773 local government areas; national total 2.10 million).

The platform: a live tool, not a static report

A no-code web application that re-runs every model on uploaded data.

Live, no-code re-runs

Upload new DHIS2 or survey files. The app checks the data and updates the results.

The LGA Priority and Archetypes screen

Filter by state or LGA to see vaccine risks, estimated missed children, rank and recommended response type. Results export to Excel.

ZARA, the platform's analyst

Zero-dose Analytics and Risk Assistant: ZARA answers simple questions in English, Hausa, Yoruba, Igbo, Pidgin and more, using only the results in the platform.

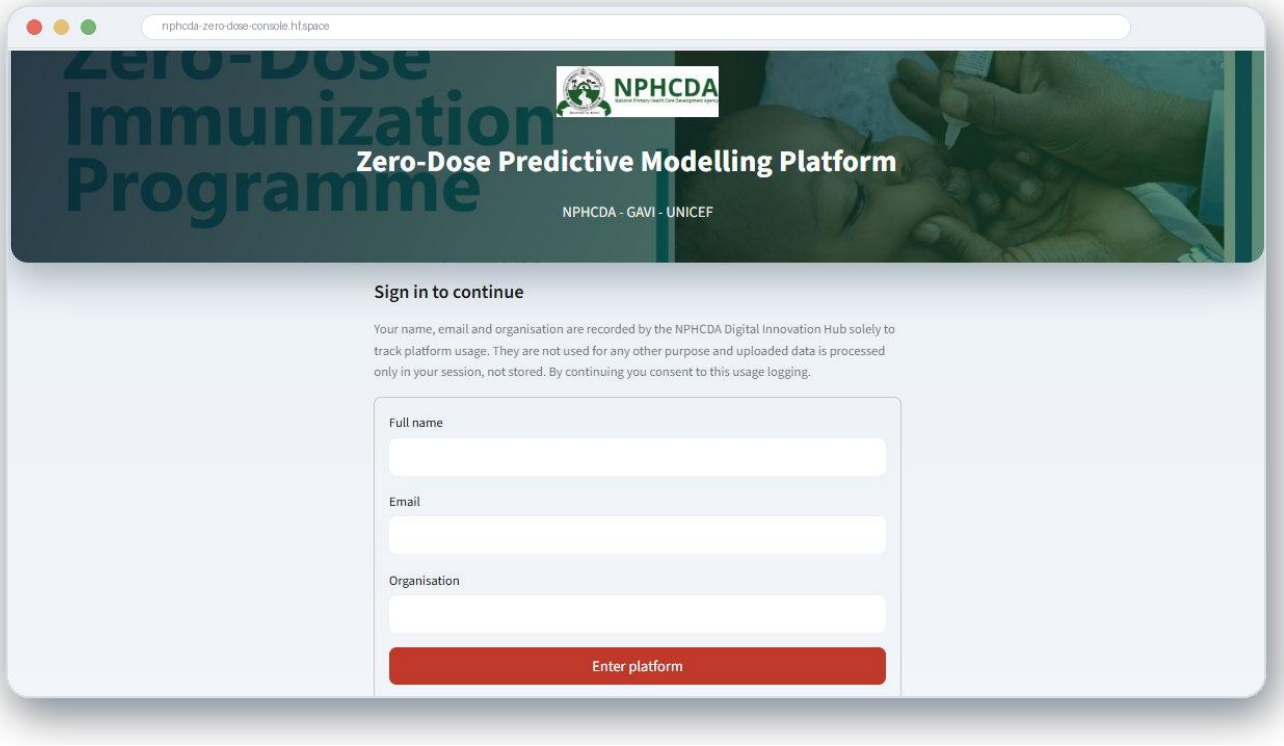
Guardrails, on every view

Each result shows the data date, reporting completeness and whether it is an estimate. This helps users avoid misreading the figures.

Live demonstration: huggingface.co/spaces/amobionovo/nphcda-zerodose

WHAT THE WEB PLATFORM ADDS

From notebooks to a live decision platform the DIH team can operate



The live platform on Hugging Face; secure login with usage tracking.

Live, real-time modelling

Runs on uploaded data, deployed on Hugging Face Docker.

Reproduces the report

National total within 0.01%; back-tested and wave-validated.

Grounded AI layer

Interpreter on every output, ZARA analyst, page-cited Q&A.

Two coverage lenses

Estimated coverage and the 2024 index, triangulated with NDHS.

Decision-ready outputs

Scorecards, maps, GAVI factsheet, policy brief, this deck.

Governance built in

Secure login and usage tracking; data stays in-session.

THE HUGGING FACE APPLICATION

How the tool supports programme discussions

It re-runs live, explains itself in plain language, and produces the documents a meeting needs.

1. **Live re-runs on new data.** Upload refreshed files; the app validates the structure and recomputes only what changed. No code change is needed.

2. **ZARA, a grounded analyst.** Plain-language and multilingual Q&A (English, French, Hausa, Yoruba, Igbo, Pidgin, Swahili and more). It reads the live outputs and is instructed never to invent numbers.

3. **One-click factsheet and policy brief.** A GAVI-style factsheet and a Word policy brief regenerate from the same live outputs, so the documents never drift apart.

4. **Governance built in.** Secure login, data stays in-session, and the load-validate-run-review-approve-release sign-off cycle for the DIH.

ZARA is grounded on the live results and uploaded documents (with page citations). One source of truth feeds the report, factsheet, brief and deck.

Recommendations from the modelling team

1. **Act first where burden and deprivation meet**
   Begin with Sokoto, Kebbi, Zamfara and the highest-burden LGAs. About 151 LGAs reach half of missed children; the 107 LGAs ranked in the top 155 by both methods come first.

2. **Match packages to each LGA's measured barriers**
   Start from the state package and add outreach, security-sensitive access, maternal-care or nutrition integration, demand support or defaulter tracing where an LGA's barriers call for it; validate locally.

3. **Adopt the platform for routine microplanning**
   Use monthly DHIS2 data to update the priority list, flag data problems and support Gavi 6.0 targeting.

4. **The Digital Innovation Hub leads from here**
   The DIH should lead routine use of the platform, with continued collaboration for upgrades and expansion to nutrition and child health.

The results have been checked against the newest national survey and can support Gavi 6.0 targeting.

Acronyms and abbreviations (1 of 2)

Every acronym used in this deck, spelled out.

ACLED	Armed Conflict Location and Event Data
ACSM	Advocacy, Communication and Social Mobilization
AFP	Acute flaccid paralysis
ANC	Antenatal care
BCG	Bacille Calmette-Guerin (tuberculosis vaccine)
CHI	Composite Hesitancy Index
cVDPV	Circulating vaccine-derived poliovirus
DHIS2	District Health Information System, version 2
DHS	Demographic and Health Survey
DIH	Digital Innovation Hub (NPHCDA)
DTP1 / Penta1	First dose of the pentavalent vaccine
FDR	False discovery rate
GBD	Global Burden of Disease study (IHME)
Getis-Ord Gi*	Local spatial hotspot statistic
GRID3	Geo-Referenced Infrastructure and Demographic Data for Development
HTR	Hard-to-reach
IDSR	Integrated Disease Surveillance and Response
IHME	Institute for Health Metrics and Evaluation
ITN	Insecticide-treated net
LASSO	Least absolute shrinkage and selection operator

Reference. CIDRE and Quantium Insights LLC, in technical support of NPHCDA.

Acronyms and abbreviations (2 of 2)

Every acronym used in this deck, spelled out.

LGA	Local government area
MCV1 / Measles1	First dose of measles-containing vaccine
Measles2	Second dose of measles-containing vaccine
MGWR	Multiscale geographically weighted regression
NCDC	Nigeria Centre for Disease Control
NDHS	Nigeria Demographic and Health Survey
NmDHS	Nigeria mini Demographic and Health Survey
NPC	National Population Commission
NPHCDA	National Primary Health Care Development Agency
OLS	Ordinary least squares regression
SAE	Small-area estimation
Penta3	Third dose of the pentavalent vaccine
R-hat	Gelman-Rubin convergence diagnostic
RI	Routine immunization
RMNCH	Reproductive, maternal, newborn and child health
RR	Relative risk
RWI	Relative Wealth Index (Meta)
SIA	Supplementary immunization activity
UNICEF	United Nations Children's Fund
VPD	Vaccine-preventable disease
ZARA	Zero-dose Analytics and Risk Assistant
BYM2	Besag-York-Mollie spatial model, version 2

Reference. CIDRE and Quantium Insights LLC, in technical support of NPHCDA.